Decentralized Exchanges and Lending

Blockchain in Traditional Finance Service



Cross-Border Transactions, Clearing and Settlements



Point of Sales (PoS) Systems



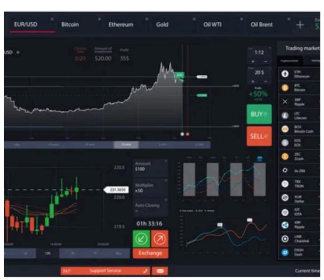
Insurance



Lending and Borrowing



Smart Bonds / Hedge Funds



Trading Platforms



Bookkeeping and Auditing



Digital Identity, Credit Reports

Blockchain in Traditional Finance Service

- More Efficient with Automatic Verification Process
- More Secure with Distributed Liability for Funds
- Higher Potential for Increased Value on Digital Assets
- Lower Associated Costs

Decentralized Finance



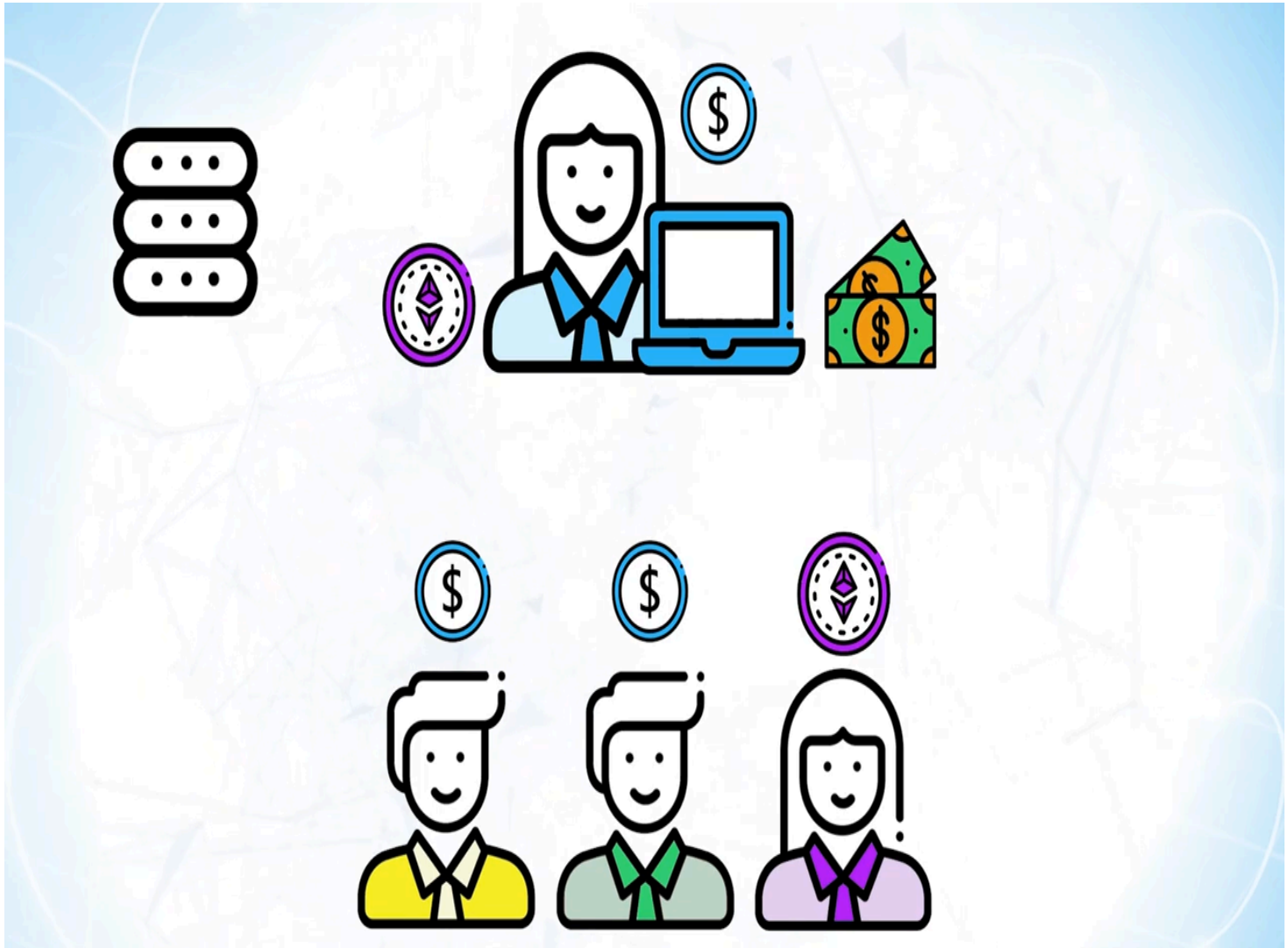
Decentralized Finance



Most Successful Decentralized Application:



Initial Coin Offering (ICO)



What is an ICO?

What is an ICO?

Fundraising Tool

- a new method of funding for start-ups
- a potential replacement of IPO?

Tokenization

- new digital tokens or coins are issued
- utility tokens or stocks (governance)?

What is an ICO?

Initial Coin Offering — The crypto equivalent of an IPO

An ICO is an early-stage fundraising mechanism for blockchain projects.

The project team issues new tokens, and investors purchase them using established cryptocurrencies like ETH or BTC. The team receives development capital; investors receive tokens — hoping for future appreciation or utility value.

Think of it as: ICO ≈ IPO, but with almost no regulatory barriers, global participation, and blazing speed.

	IPO	ICO
Regulation	Strict	Minimal
Barrier	High	Near zero
Audience	Institutions	Global retail
Timeline	6-12 months	Days-weeks

A Brief History of ICOs



At the 2017 peak, EOS alone raised ~$4.1 billion through its year-long ICO.

ICO Structure: Token Design

Token types, allocation, and tokenomics

Utility Token

Grants access to a product or service within the platform. E.g. ETH for gas, FIL for storage, UNI for governance.

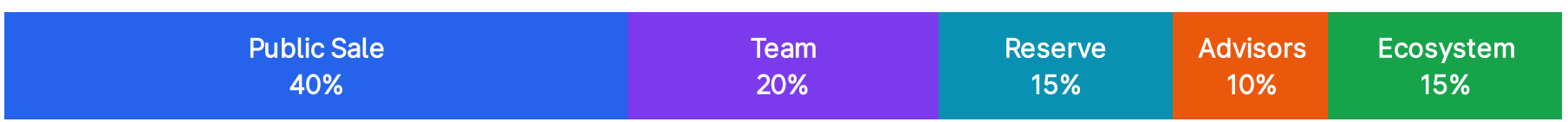
Security Token

Represents equity, dividends, or asset shares. Subject to securities law; requires compliant issuance (STO).

Payment Token

Functions as a medium of exchange. E.g. BTC, LTC. Treated like currency; some jurisdictions regulate separately.

Typical Token Allocation



Key Tokenomics Parameters

Total Supply

Max tokens ever created (e.g. 1 billion)

Hard Cap

Max fundraising target (e.g. $30M)

Soft Cap

Min fundraising goal (refund if not met)

Token Price

Price per token (e.g. 1 ETH = 5000 TKN)

Vesting

Lock-up & release schedule (prevents team dumping)

The ICO Process: Six Stages

From concept to exchange listing

1 Concept & Whitepaper

Draft whitepaper: technical design, tokenomics, team bios, and roadmap

2 Smart Contract

Develop & deploy the token contract (usually ERC-20); third-party audit

3 Marketing & Community

Build community (Telegram / Discord), KOL outreach, AMAs, airdrops

4 Pre-sale (Private)

Private round for institutions & early backers; discounted price with lock-up

5 Public Sale

Open sale to the public within a set time window; first-come or whitelist

6 Exchange Listing

Token lists on exchanges (CEX / DEX); secondary market trading begins

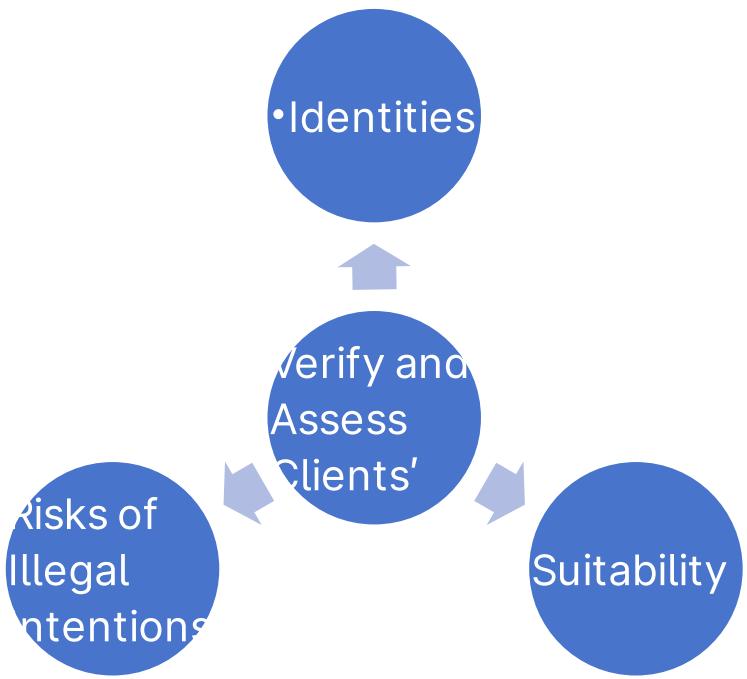
Typical Timeline: Whitepaper (1-3 mo) → Contract Dev & Audit (1-2 mo) → Community Building (1-3 mo) → Private Sale (1-2 wk) → Public Sale (1 day - 1 mo) → Listing (1-3 mo after sale)

End-to-end: typically 3-9 months — significantly faster than a traditional IPO (6-18 months).

Know Your Customer (KYC)

bank regulations and anti-money laundering regulations





What is KYC?

Know Your Customer — Definition & Purpose

Definition

KYC (Know Your Customer) is a set of mandatory procedures that financial institutions use to **verify the identity** of their clients before and during business relationships.

It is a core component of AML (Anti-Money Laundering) and CFT (Combating the Financing of Terrorism) compliance frameworks.

01 **Prevent Fraud**
Confirm customers are who they claim to be

02 **Block Money Laundering**
Detect and report suspicious financial activity

03 **Counter Terrorism**
Prevent funds from reaching terrorist organizations

Regulatory Context: Required by laws such as the U.S. Bank Secrecy Act (BSA), EU's 4th/5th/6th Anti-Money Laundering Directives (AMLD), and FATF recommendations. Non-compliance can result in heavy fines (millions to billions) and criminal prosecution.

The KYC Process: Three Pillars

1 CIP

Customer Identification Program

- Collect full legal name, date of birth, address, ID number
- Verify via government-issued ID (passport, driver's license)
- Cross-check against government databases

2 CDD

Customer Due Diligence

- Assess the nature and purpose of the relationship
- Identify Beneficial Owners (>25% ownership)
- Classify risk level: low / medium / high

3 EDD

Enhanced Due Diligence

- For high-risk customers: PEPs, sanctioned countries
- Source of funds / source of wealth investigation
- Senior management approval required

Low Risk → CIP + CDD sufficient | High Risk → CIP + CDD + EDD required

KYC in Crypto & DeFi

The tension between compliance and decentralization

CeFi — Centralized Exchanges

Full KYC is mandatory

Exchanges like Binance, Coinbase, Kraken require:

- Government-issued photo ID
- Proof of address (utility bill)
- Selfie / liveness check
- Source of funds declaration (large amounts)

Without KYC: limited or no withdrawals

DeFi — Decentralized Protocols

Mostly no KYC today

Protocols like Uniswap, Aave are permissionless:

- Connect wallet → trade immediately
- No identity verification required
- Pseudonymous by design

Emerging trend:
On-chain KYC solutions (e.g. SBTs, zkKYC) allow compliance without revealing full identity

Trend: Regulators worldwide (FATF Travel Rule, EU MiCA) are pushing DeFi towards KYC. The balance between privacy and compliance is the central challenge for Web3.

Famous ICOs by amount raised

Block.one (EOS)	$4.0 billion
Telegram Open Network	$1.7 billion
Dragon Coin	$320 million
Huobi	$300 million
HDac	$258 million
Filecoin	$257 million
Tezos	$232 million
Sirin Labs	$158 million
Bancor	$152 million
Bankera	$151 million

Block.one's $4 billion ICO on May 2018

CNBC MARKETS INVESTING TECH

A blockchain start-up just raised $4 billion without a live product

- Cayman Islands start-up Block.one raised $4 billion as of Thursday, eclipsing the world's biggest initial public offerings this year and more than doubling the next biggest offering of that type.
- But its product doesn't fully launch until June 1 and investors don't know how the majority of that capital will be used.
- "They have set the bar very high for themselves in terms of delivery expectations," says William Mougayar, managing partner at JM3 Capital and author of "The Business Blockchain."



Brendan Blumer, CEO, Block.One Source: Block.one

Criteria for “Success”

- Liquidity: Trading Volume

- Investors clearly respond to aspects of ICO design
  - Utility Features
  - Reduction of Asymmetric Information
  - Background of Founder

Method of Distribution

- Had a pre-sale 45%
- Dynamic pricing 34%
- Auction pricing 5%

- Accepted ETH 66%
- Accepted BTC 41%
- Accepted LTC 9%
- Accepted USD 10%
- Accepted EUR 3%

Bonding and Disclosure

- Twitter page 97%
- Website 92%
- Telegram group 83%
- White paper 81%
- Source code posted on Github 66%
- Lock-up period for founder 36%
- VC-backed 11%

Most tokens reside on Ethereum Thanks to ERC-20 Token!

Smart Contract Audit Services

Who needs the Smart Contracts Audit?



Smart Contracts Developers



Owners & Managers of ICO Startups



Product Owners of Decentralized Apps



Anybody Who Needs Trust of Stakeholders & Investors

Why Security Auditing?

The hard lessons from the ICO era

ICO Era Problems (2017-2018)

Thousands of ICOs launched with unaudited smart contracts, leading to:

- Funds locked forever due to bugs (Parity: $300M frozen)
- Reentrancy exploits (The DAO: $60M drained)
- Backdoors enabling rug pulls by developers
- Integer overflow / underflow vulnerabilities

These disasters made auditing the industry standard.

Auditing as the Answer

Security auditing became the Web3 trust layer:

- Pre-launch code review to catch vulnerabilities
- Public audit reports build investor confidence
- Exchanges now require audits before listing
- Investors check "Is it audited?" as first due diligence

No audit = No trust = No investment.

From ICO to Today: The Rise of Auditing



The Audit Process

How a professional smart contract audit works

1 Scoping

Define which contracts, commit hash, and threat model. Agree on timeline and cost ($5K–$500K+).

2 Automated Analysis

Run static analyzers (Slither, Mythril), fuzzers (Echidna), and symbolic execution tools.

3 Manual Review

Line-by-line expert review: business logic, edge cases, economic attacks, gas optimization.

4 Findings Report

Classify issues by severity (Critical / High / Medium / Low / Info) with fix recommendations.

5 Remediation & Re-audit

Dev team fixes issues; auditor verifies fixes. Final report published for transparency.

Key Tools & Leading Firms

Automated Tools

Slither (static) • Mythril (symbolic) • Echidna (fuzzing) • Foundry • Certora (formal verification)

Top Audit Firms

OpenZeppelin • Trail of Bits • CertiK • Consensys Diligence • Halborn • Spearbit • Code4rena (competitive)

What Gets Audited?

Smart contract vulnerabilities that auditors hunt for

Token Contracts

ERC-20 / ERC-721 / ERC-1155 token logic, minting, transfer, and access control

DeFi Protocols

DEX, lending, staking — price oracle manipulation, flash loan attacks, liquidity drains

Bridge & Cross-chain

Cross-chain messaging, validator sets, replay attacks — highest-value targets

NFT & GameFi

Minting logic, metadata handling, randomness, and marketplace interactions

Top Smart Contract Vulnerabilities (OWASP / SWC)

Vulnerability	Description	Severity	Notable Exploit
Reentrancy	Recursive calls drain funds before state update	Critical	The DAO ($60M)
Oracle Manipulation	Attacker corrupts price feed to extract value	Critical	Mango Mkts ($114M)
Flash Loan Attack	Uncollateralized loan exploits pricing logic	High	bZx ($8M)
Access Control	Missing permission checks on privileged functions	Critical	Wormhole ($326M)

Total DeFi hack losses exceed $6 billion since 2020 — the majority through smart contract exploits.

Two Major Regulatory Issues with ICOs

Are ICOs covered by the securities laws?

- Of which country?
- Why is this important?

What are the tax consequences?

- Currency?
- Assets?

ICO Risks & Red Flags

Why 90%+ of ICO projects ultimately fail or turn out to be scams

Common Risks

Rug Pull / Exit Scam
Team disappears with the funds; investors lose everything

No Product / Vaporware
The product described in the whitepaper is never built — concept only

Token Dump
Team / insiders mass-sell tokens, crashing the price on retail holders

Regulatory Crackdown
Classified as unregistered securities by SEC or similar bodies; project shut down

Red Flags Checklist

- ⚠ Anonymous team with no verifiable identity
- ⚠ Whitepaper is vague, lacks detail, or plagiarized
- ⚠ Unrealistic promises ("guaranteed 100× returns")
- ⚠ Smart contract is unaudited or closed-source
- ⚠ Team allocation exceeds 30% of total supply
- ⚠ No vesting / lock-up schedule for insiders
- ⚠ Community filled with bots, no genuine engagement

80% of ICOs classified as scams (Satis, 2018)

$100M+ lost in the Bitconnect fraud

46% of 2017 ICOs failed within 4 months

Token Lockup (Vesting Periods)

tokens are not part of the circulating

often set as 1 or 2 years after the ICO

period of time in which tokens

Purpose

- to maintain a stable long-term value of a particular asset
- to prevent the holders of big bags to sell their tokens all at once
- to bring an extra level of confidence to the potential participants

Warning:
ICO is illegal in China

Illegal Fundraising

Decentralized Finance



What is an Airdrop & Why Do It?

Free token distribution as a growth and decentralization strategy

An airdrop is the free distribution of tokens to a large number of wallet addresses.

Projects send tokens directly to wallets — either unconditionally or as a reward for past on-chain activity. It is one of the most powerful user acquisition tools in Web3.

01 **User Acquisition** Bootstrap an active community from day one

02 **Decentralize Governance** Distribute voting power widely for DAO legitimacy

03 **Reward Early Users** Incentivize loyalty & retroactive contributions

04 **Marketing & Hype** Generate buzz, media coverage, and social engagement

Types of Airdrops

Standard Airdrop

Anyone can claim by signing up, following social accounts, or joining a community.

Holder Airdrop

Tokens sent to holders of a specific asset. E.g. ETH holders receive a new token.

Retroactive Airdrop

Rewards past on-chain users. The most valuable type — e.g. Uniswap, Arbitrum.

Bounty Airdrop

Earned by completing tasks: tweets, referrals, bug reports, content creation.

Retroactive airdrops have created billions in value — Uniswap's $UNI airdrop alone was worth ~$6,400 per wallet at peak.

Airdrop Process: Step by Step

How a project designs and executes a token airdrop

1 Define Objective

Goal: growth, governance decentralization, or reward? This determines the type and criteria.

2 Set Eligibility Criteria

Snapshot date, min activity thresholds, exclude bots/Sybils, geographic restrictions.

3 Snapshot & Data Collection

Take on-chain snapshot at a specific block. Compile eligible addresses & compute allocations.

4 Token Allocation

Design distribution tiers. More activity = more tokens. Typical: 5-15% of total supply for airdrop.

5 Claim / Distribute

Merkle tree claim contract (user pays gas) or direct distribution (project pays gas).

6 Post-Airdrop Strategy

Staking incentives, governance activation, vesting for large recipients to reduce sell pressure.

Technical: How Claims Work (Merkle Tree)

1. Project builds a Merkle tree from all eligible (address, amount) pairs and publishes the Merkle root on-chain.
2. Each user calls claim(amount, merkleProof) — the contract verifies the proof against the root. Only valid claimants succeed.
3. This is gas-efficient: no need to store all addresses on-chain. The user pays gas to claim, reducing project cost.

Famous Airdrop Case Studies

Uniswap ($UNI)

Sep 2020

Allocation: 15% of supply (150M UNI)

Per Wallet: 400 UNI (~$1,200 at launch, ~$6,400 at peak)

Eligibility Criteria

Anyone who used Uniswap V1 or V2 before Sep 1, 2020

Impact & Legacy

Set the standard for retroactive airdrops. Created massive community loyalty overnight.

Arbitrum ($ARB)

Mar 2023

Allocation: 11.6% of supply (1.16B ARB)

Per Wallet: 625 – 10,250 ARB (tiered by activity)

Eligibility Criteria

Bridge users, tx count, time span, liquidity, used >2 dApps, etc.

Impact & Legacy

Point-based system with anti-Sybil filters. $120B+ claimed. Became the template for L2 airdrops.

Optimism ($OP)

Jun 2022

Allocation: 5% of supply (initial), 19% total over time

Per Wallet: Varied by 6 overlap criteria

Eligibility Criteria

OP mainnet users, repeat users, DAO voters, multi-sig signers, Gitcoin donors

Impact & Legacy

Multi-round airdrop design. Rewarded governance participation, not just transactions.

Trend: airdrops are evolving from simple giveaways → activity-weighted, Sybil-resistant, multi-round distributions.

Impact of Airdrop

Revolution!!!

- Users should be the beneficiaries of the business growth
- Data Ownership!

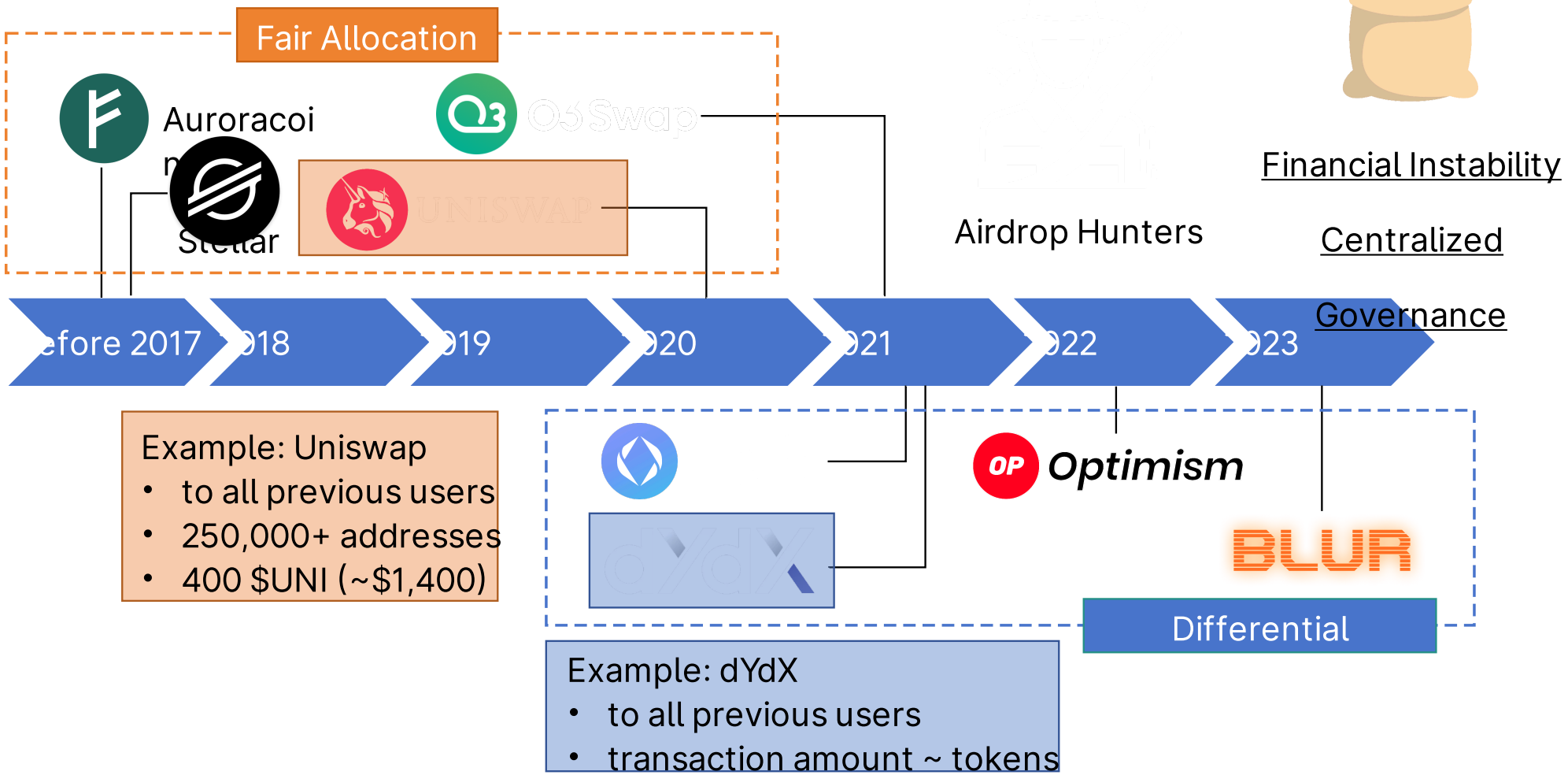
Negative Impacts

- Tax Liability Issues: Income or Capital Gains?
- Airdrop Hunters for Profits

Case: $SOS

- Dec 24, 2021
- airdrop to people who have spent money on OpenSea transactions
- from OpenDAO
- the project isn't affiliated with OpenSea
- almost 200,000 wallets have claimed $SOS tokens
- up more than 1,000% in just two days
- the contract has a market cap of over $200 million

The Development of Airdrop



The Problem: Airdrop Hunters

How hunters exploit airdrops and why protocols must evolve

What Hunters Do

- **Sybil Farming** — create 100s-1000s of wallets
- **Minimum Activity** — do the bare minimum to qualify
- **Script Automation** — bots execute identical txn patterns
- **Instant Dump** — sell all tokens immediately at listing

One farmer with 1,000 wallets can extract millions in tokens meant for genuine users.

Damage to Protocols

- **Financial Instability** — massive sell pressure crashes token price
- **Centralized Governance** — farmers accumulate disproportionate voting power
- **Community Dilution** — real users get smaller allocations
- **Reputation Damage** — "unfair" airdrops create backlash

The Evolution of Airdrop Design

Fair Allocation

Before 2017 – 2020

Equal tokens to all. Simple but easily gamed.
E.g. Auroracoin, Stellar, Uniswap (400 UNI each)

→

Differential Allocation

2020 – 2022

More activity = more tokens. Tiered by usage.
E.g. dYdX, Optimism, Blur

→

Threshold Differential

2022 – Present

Min thresholds + tiers + Sybil filtering.
E.g. Paraswap, Arbitrum, LayerZero

Strategy 1: Sybil Detection & Filtering

Identify and exclude fake wallets before distribution

On-Chain Cluster Analysis

- Same funding source (CEX deposit address)
- Identical transaction patterns & timing
- Same gas-paying wallet across accounts
- Similar token holdings & NFT overlap

Case Study:
Hop Protocol flagged 10,000+ Sybil addresses using on-chain graph analysis before $HOP airdrop.

Behavioral Fingerprinting

- Transaction frequency & volume patterns
- Time-of-day activity clustering
- Sequential txn hashes (bot signatures)
- Identical smart contract call sequences

Case Study:
Connext blacklisted 30,000+ addresses by detecting identical bridging patterns across wallets.

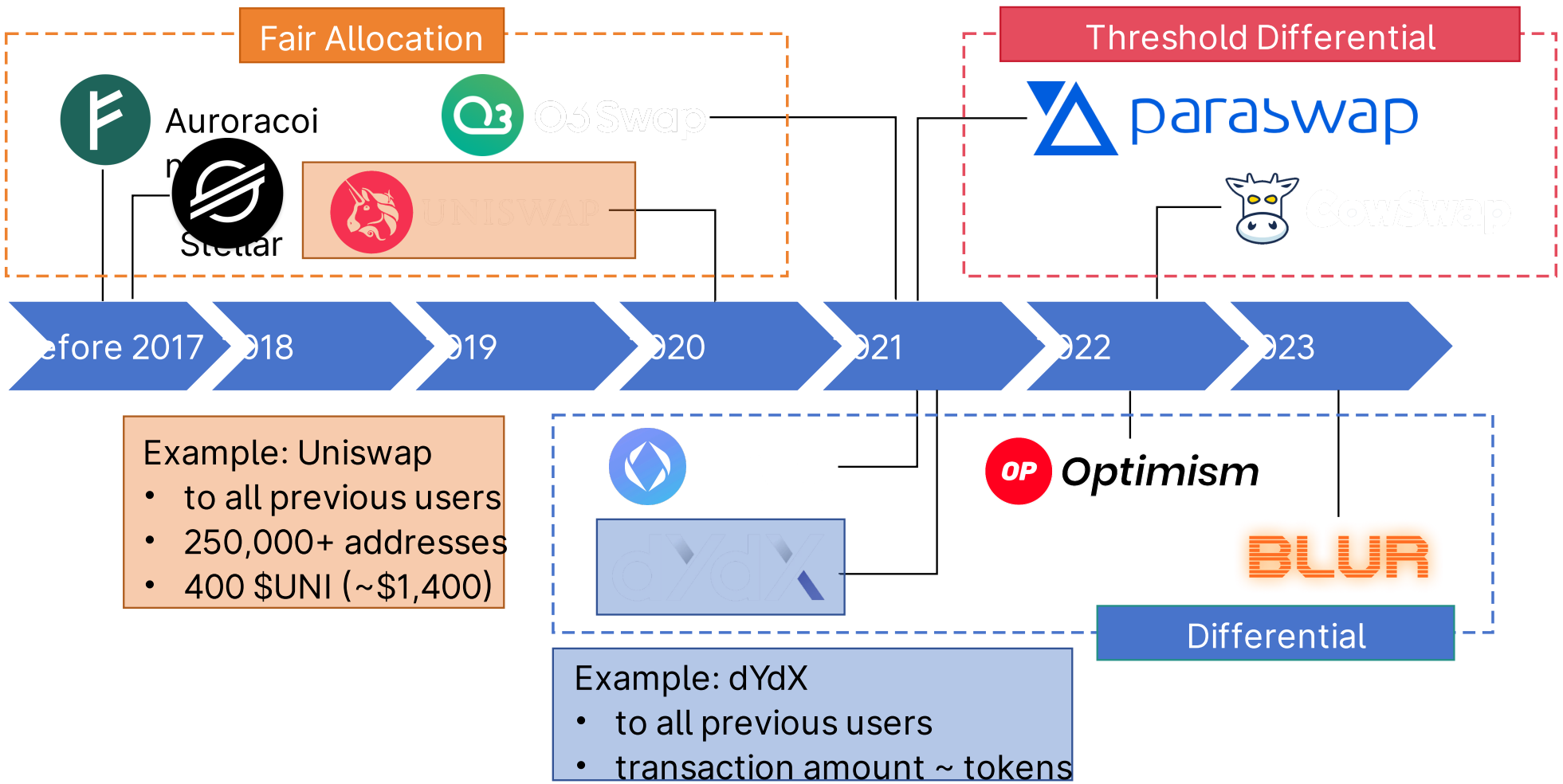
Identity Verification

- Gitcoin Passport (proof-of-humanity stamps)
- BrightID social graph verification
- ENS domain ownership as identity signal
- Worldcoin / biometric proof (controversial)

Case Study:
Optimism used Gitcoin Passport scores to boost allocations for verified unique humans.

Challenge: too aggressive filtering excludes real users; too lenient lets farmers through. Balance is key.

The Development of Airdrop



Strategy 2: Smart Allocation Design

Make farming unprofitable through mechanism design

Tiered Scoring System

Assign points for multiple dimensions: # of txns, volume, time active, unique dApps used, governance participation. Higher composite score = more tokens.

Arbitrum: 3-point min threshold, up to 15 points across 7 criteria. Top-tier users got 16× more than min.

Used by: Arbitrum, Optimism, LayerZero

Threshold + Cliff

Set a minimum activity threshold that filters out low-effort wallets. Only users who cross a meaningful usage cliff receive any allocation.

Paraswap: required min $500 volume + 5 swaps over 6 months. 80% of Sybil wallets were eliminated.

Used by: Paraswap, Blur (Season 1)

Non-Linear Rewards

Logarithmic or square-root scaling: doubling activity does NOT double rewards. This makes multi-wallet farming capital-inefficient.

If reward = √(volume): $10K across 1 wallet = 100 tokens, but $1K × 10 wallets = 10×31.6 = 316. Log curves reduce this gap.

Used by: dYdX, Blur

Multi-Round Airdrops

Don't drop everything at once. Multiple rounds reward continued engagement, not one-time farming. Later rounds can filter using data from earlier ones.

Optimism: 4 planned rounds. Round 2 rewarded governance delegation, which farmers rarely do.

Used by: Optimism, Blur (3 seasons)

ParaSwap $PSP Airdrop Rules



Threshold: Excluding Hunters

- Address Profile: < 50 txs and < minimum native token balance
- Contract Interactions: < 6 times in past 6 months
- Group Size: connected addresses > 5 and Balance < $200

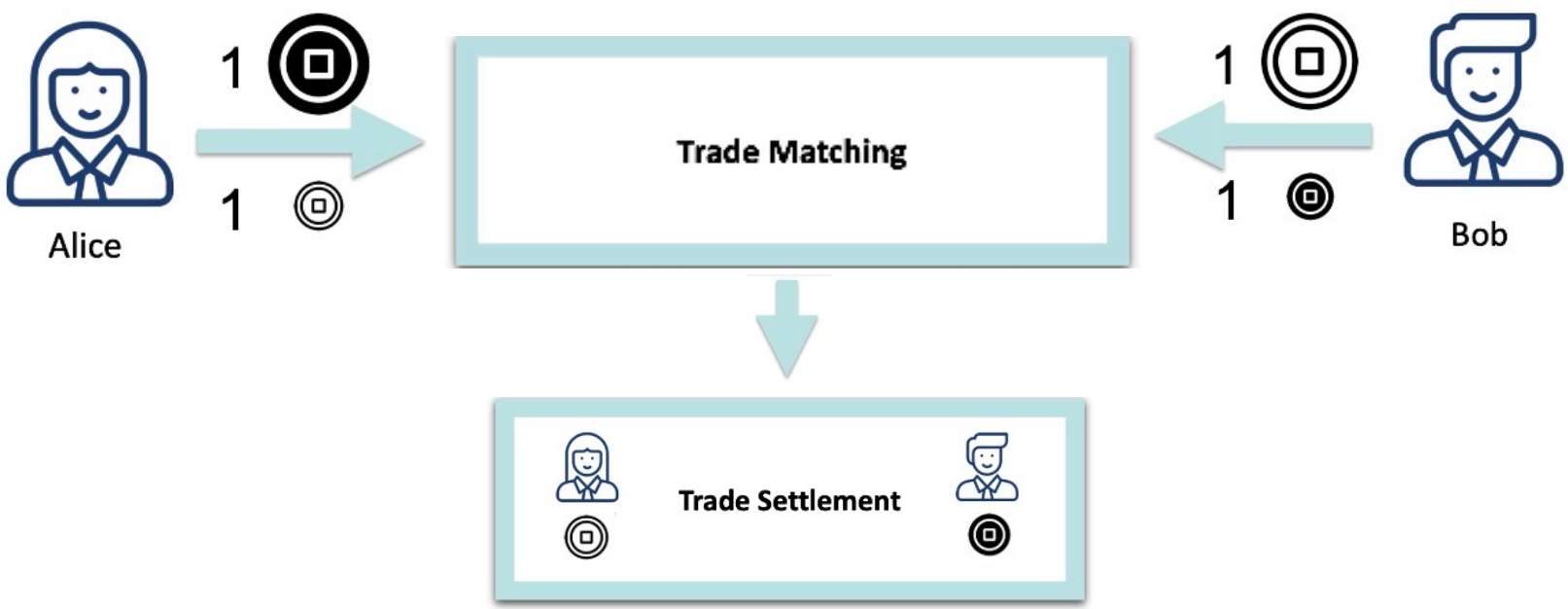
Outcome: from >1.3M “addresses” to 19,999 “active users”

Decentralized Finance



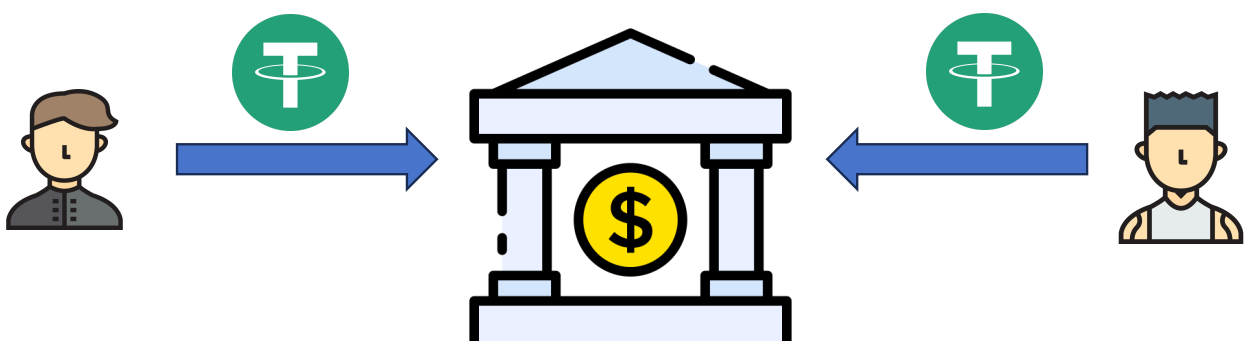
Financial Exchange

- As a trader: wants to swap token A with token B
- Need a place which lists people willing to do the reverse



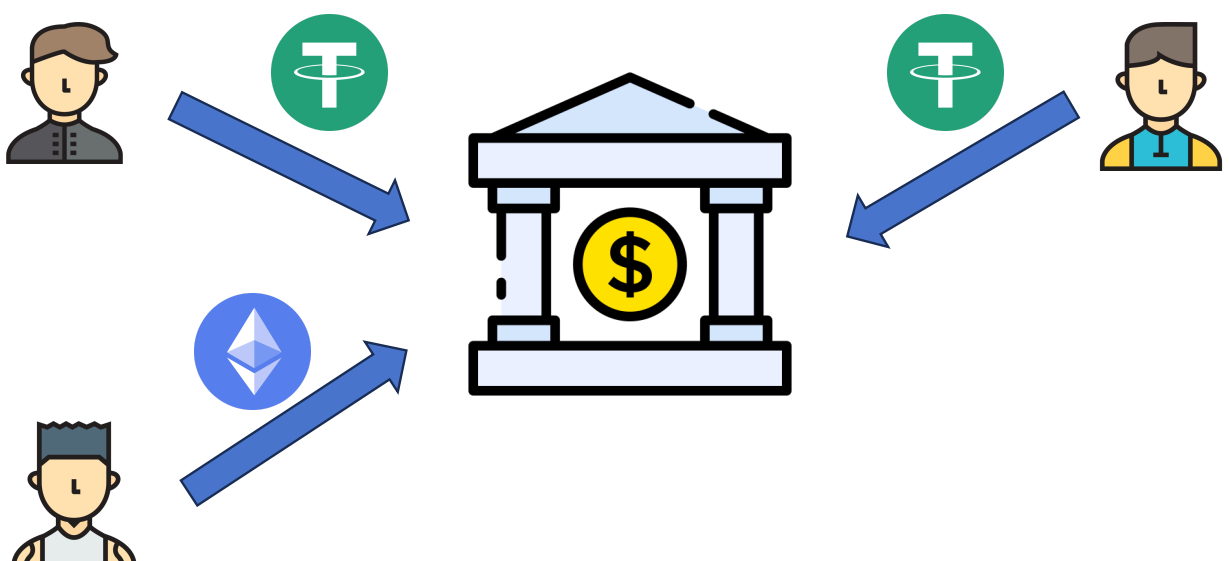
Order Book

- An order book is a list of current buy orders (also known as “bids”) and sell orders (also known as “asks”) for a specific asset.



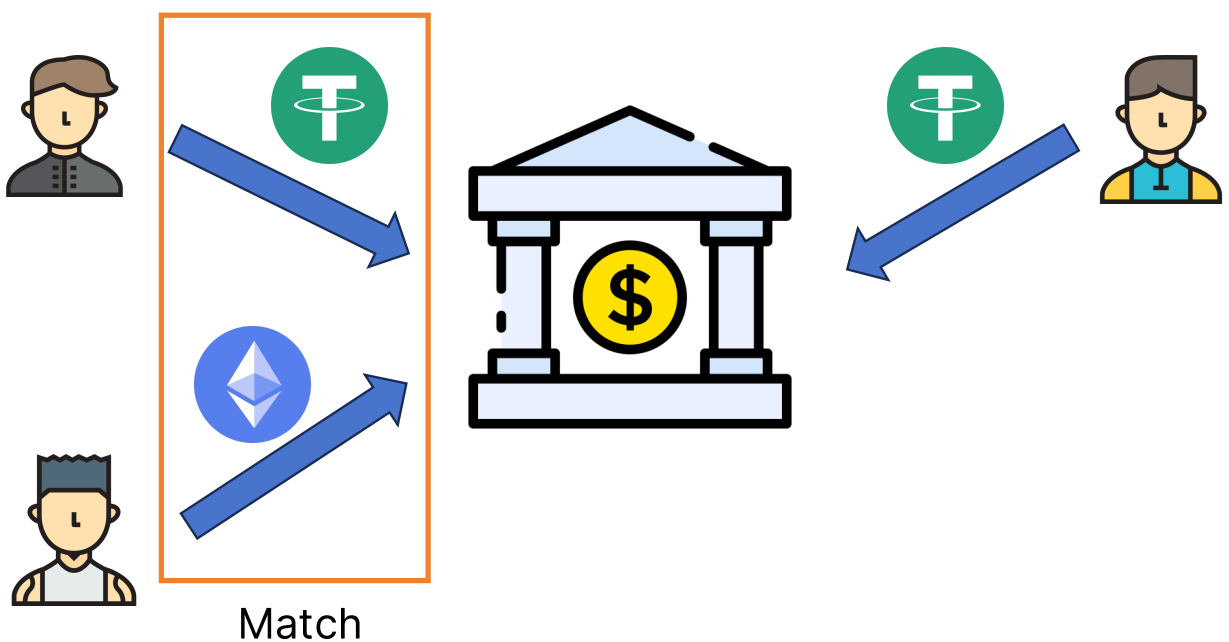
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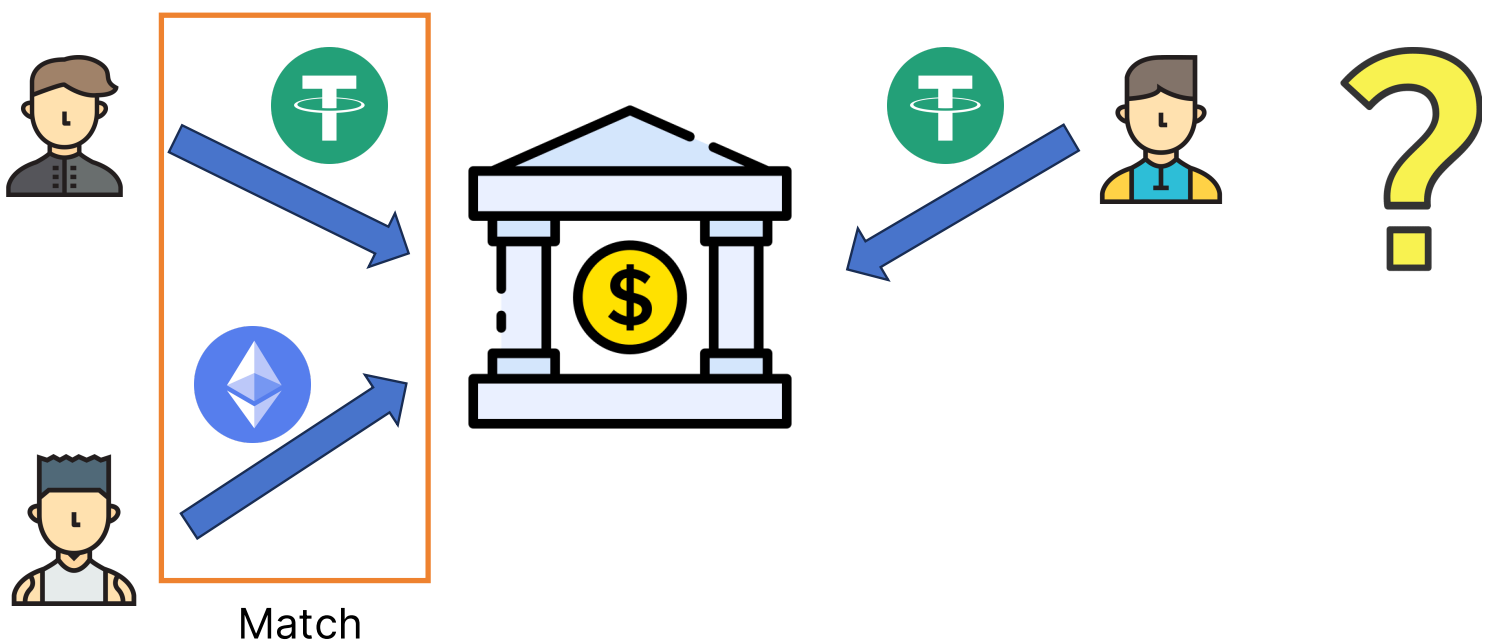
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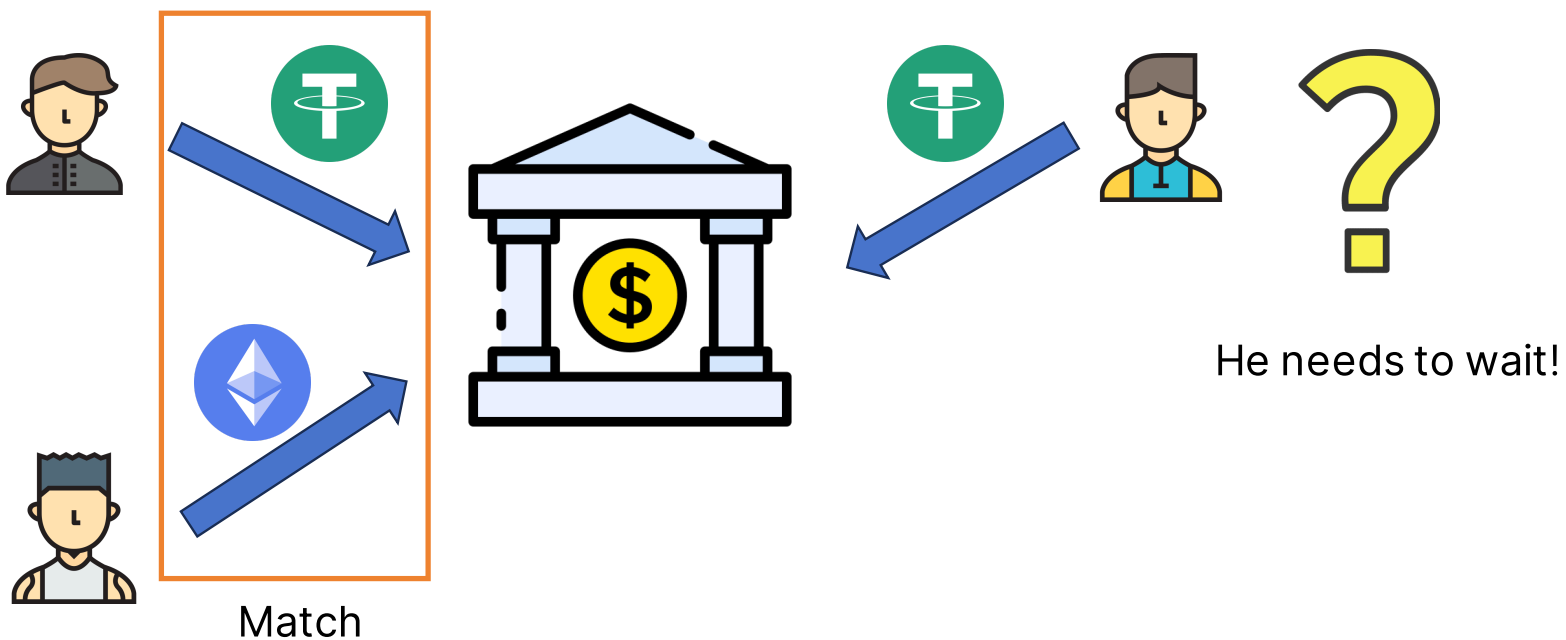
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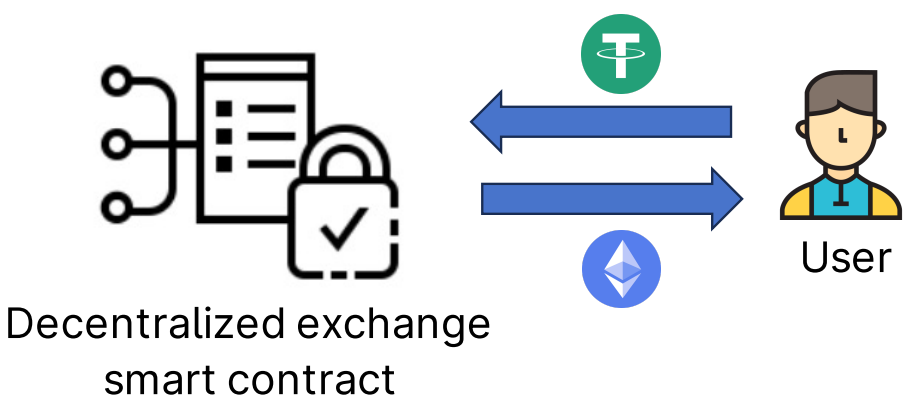
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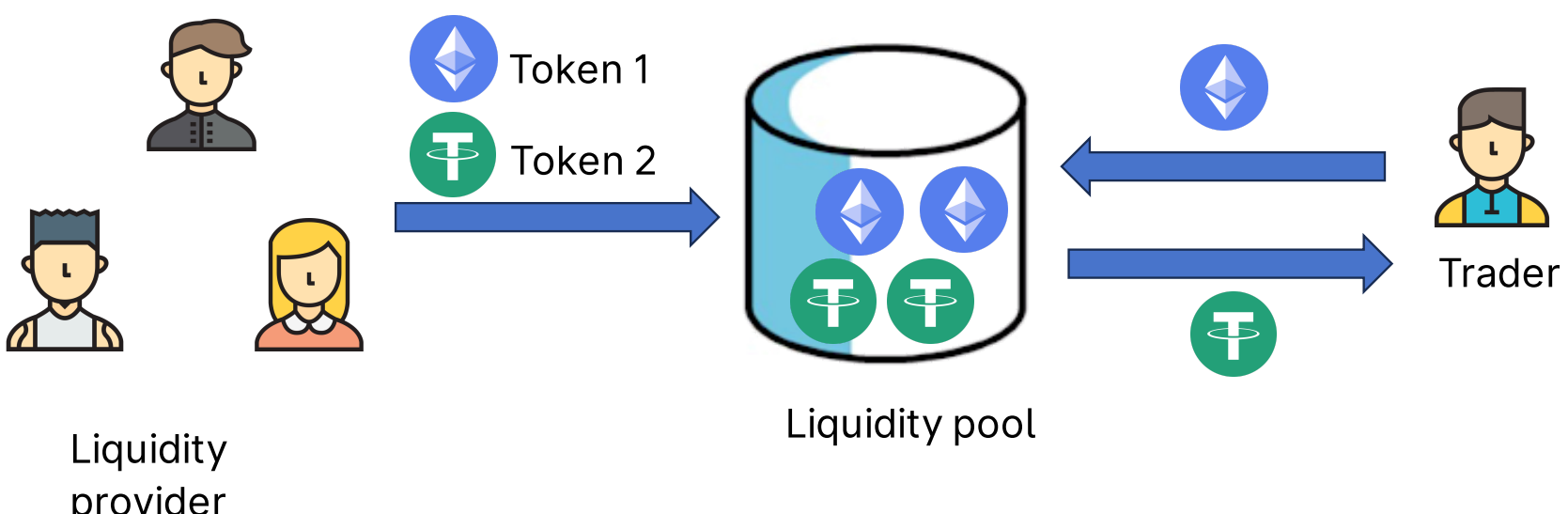
Automated Market Maker (AMM)

- Automated market makers (AMMs) are decentralized exchanges that use algorithmic “money robots” to provide liquidity for traders buying and selling crypto assets.

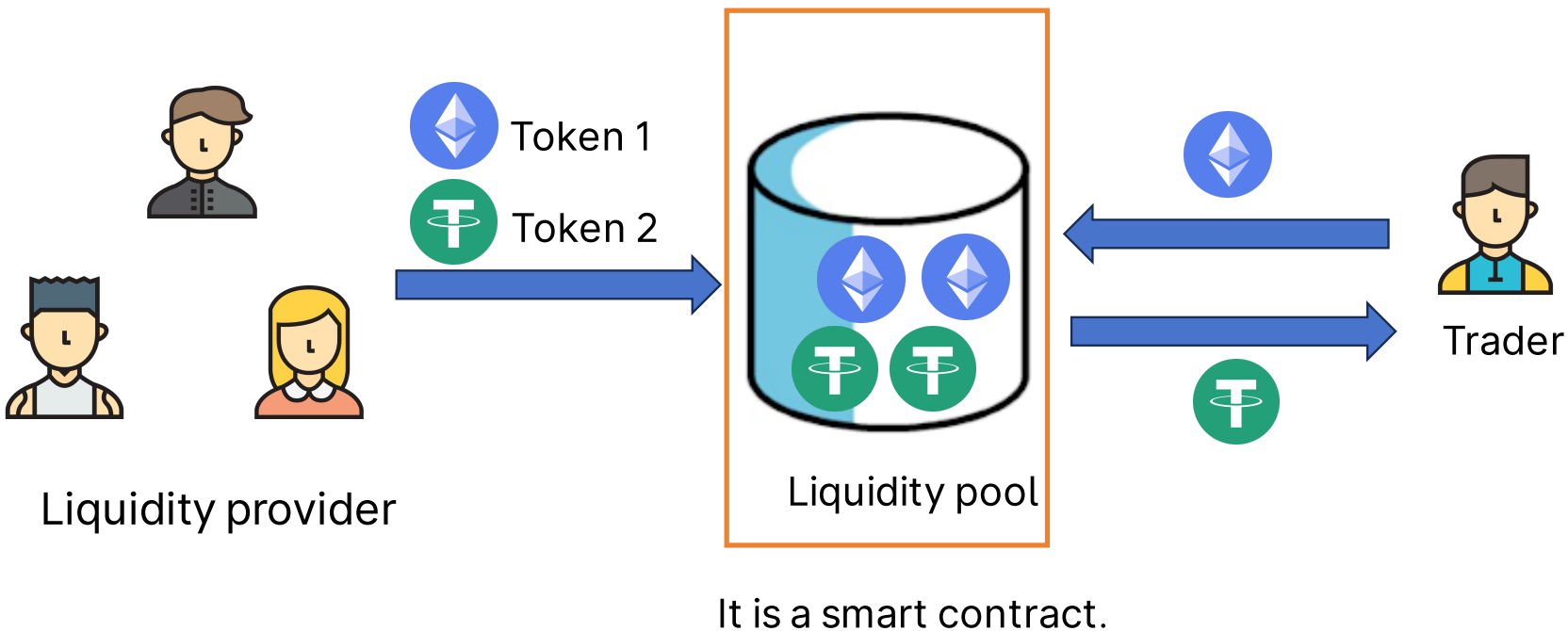


DEX System Architecture

- A liquidity provider is someone who deposits an equivalent value of two tokens into the liquidity pool within a pair. Liquidity providers take on price risk and are compensated with fees.

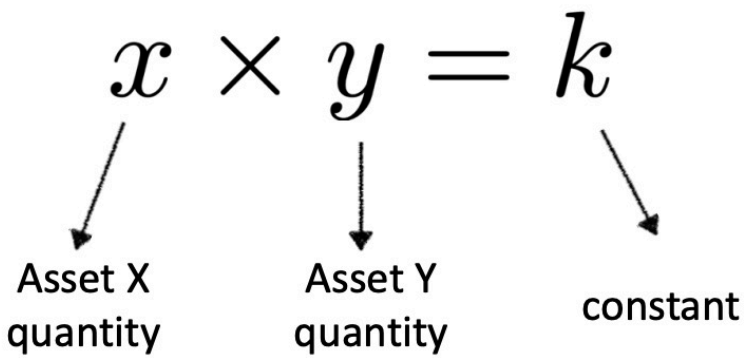


DEX System Architecture



It is a smart contract.

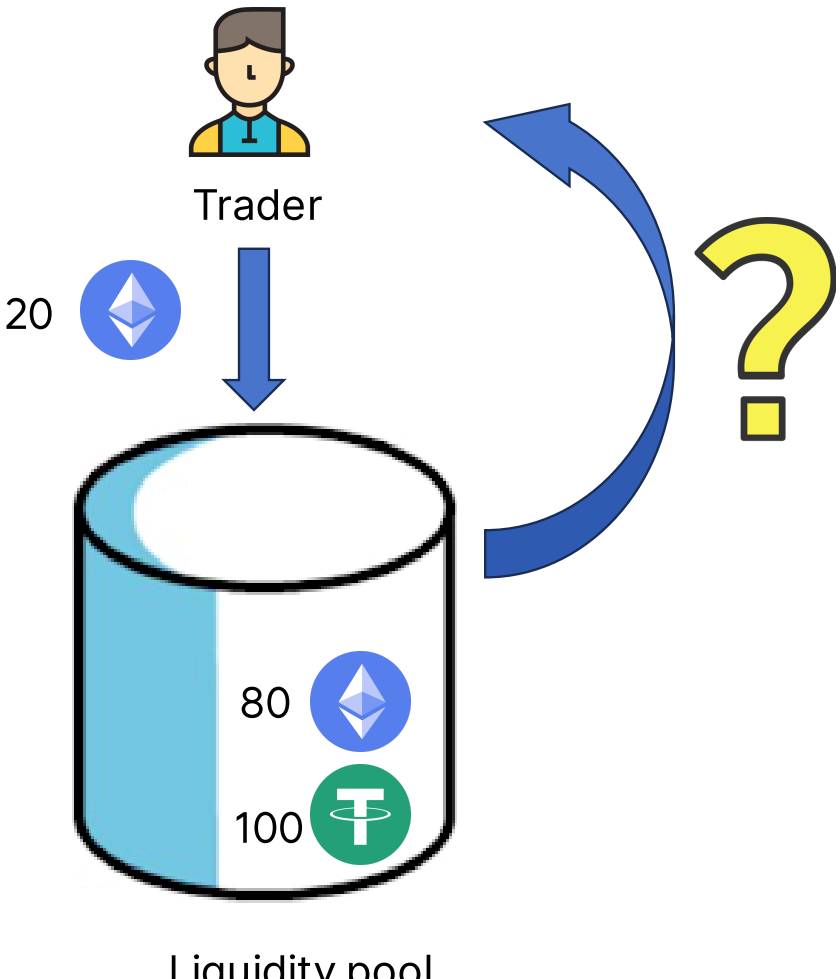
AMM



Properties:

- Instant liquidity, irrespective of the trade size.
- Purchase of asset X increases price of X and decreases the price of Y.
- Ratio of asset X and Y sets the price.
- Known as **Constant Product (CP) AMM**.

AMM



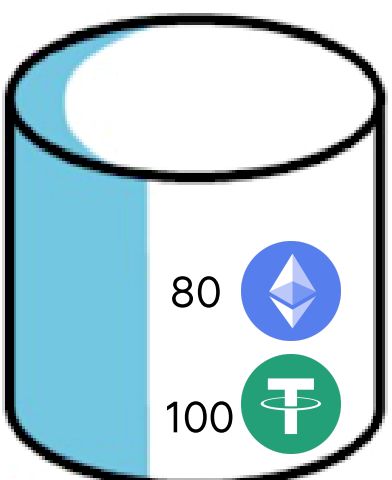
AMM

$$x \times y = k$$

Asset X quantity

Asset Y quantity

constant



Liquidity pool

80 * 100 = 8000

AMM

$$x \times y = k$$

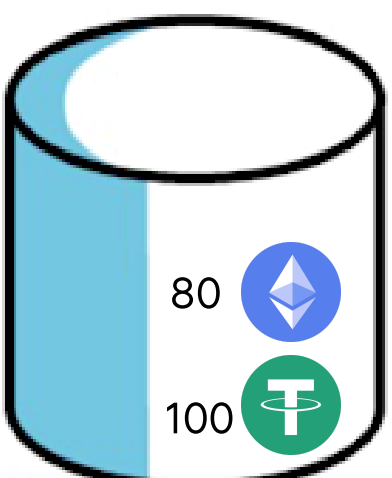
- x: Asset X quantity
- y: Asset Y quantity
- k: constant



Trader

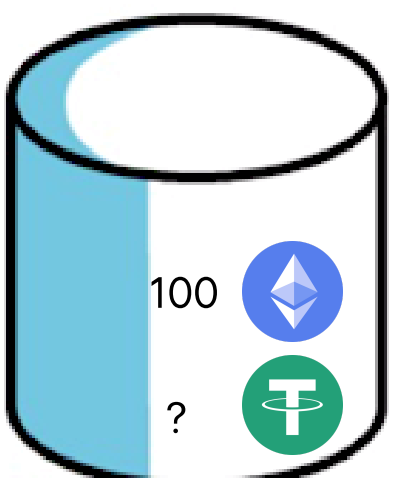
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Liquidity pool

80 * 100 = 8000



Liquidity pool

AMM

$$x \times y = k$$

Asset X quantity

Asset Y quantity

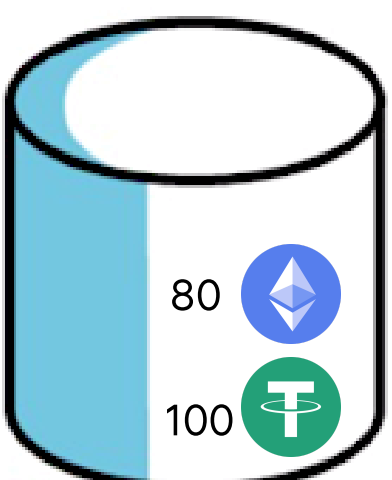
constant



Trader

20

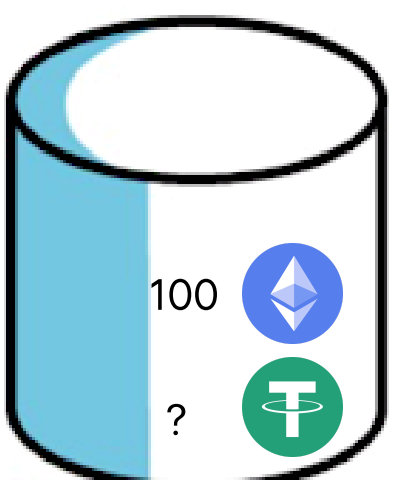




Liquidity pool

80 * 100 = 8000

(80+20) * (100-?) = 8000



Liquidity pool

AMM

$$x \times y = k$$

Asset X quantity

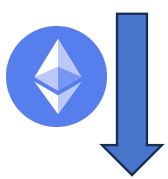
Asset Y quantity

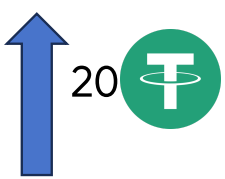
constant



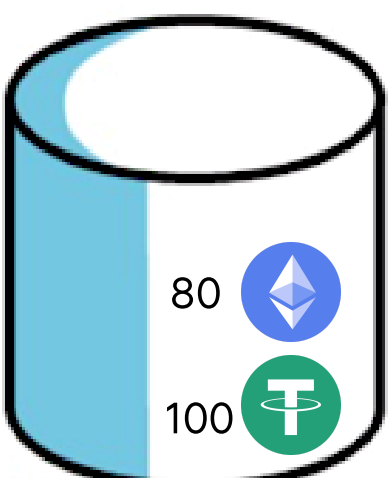
Trader

20





20



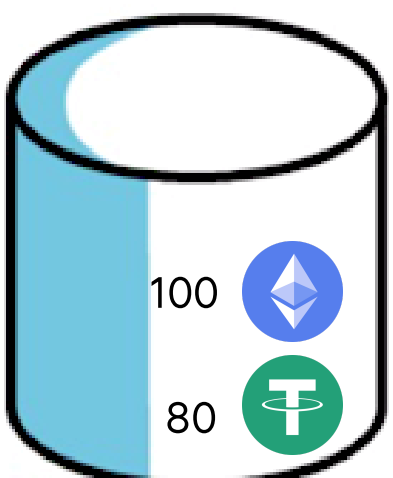
Liquidity pool

80 * 100 = 8000

(80+20) * (100-?) = 8000

100-?=80

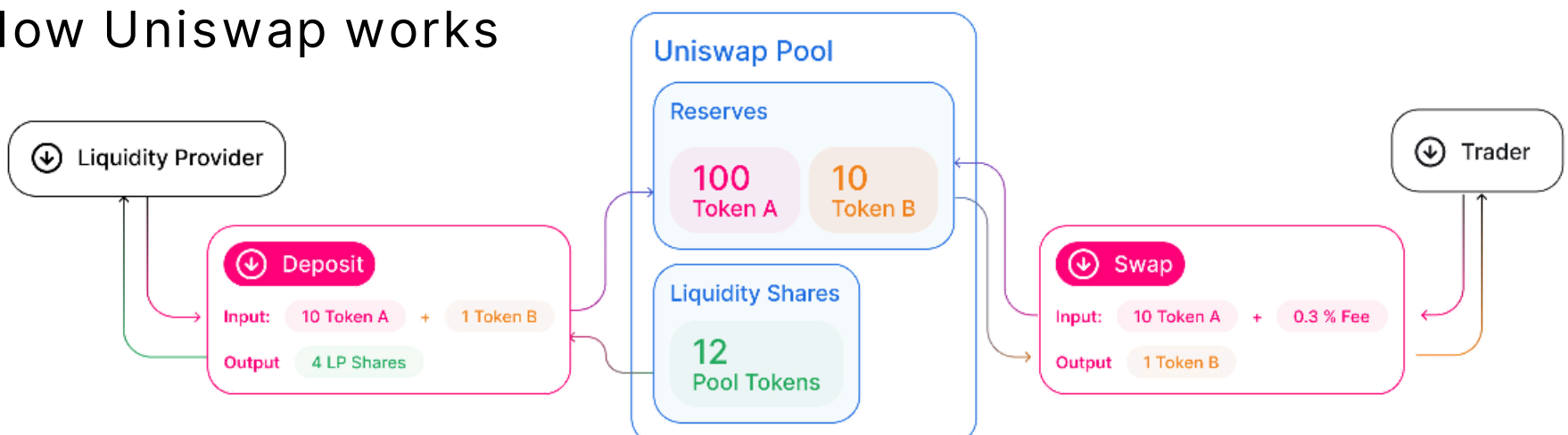
?=20



Liquidity pool

AMM Example

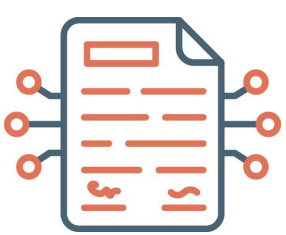
- Uniswap is the largest DEX operating on the Ethereum blockchain. It allows users anywhere in the world to trade crypto without an intermediary.
- How Uniswap works



AMM Example

- Uniswap is the largest DEX operating on the Ethereum blockchain. It allows users anywhere in the world to trade crypto without an intermediary.

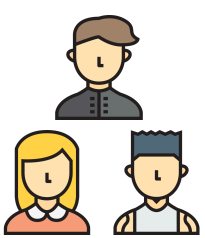
- How users take part in Uniswap?



Create new markets for exchanging new pairs of cryptos.



Swap cryptos



Liquidity provider



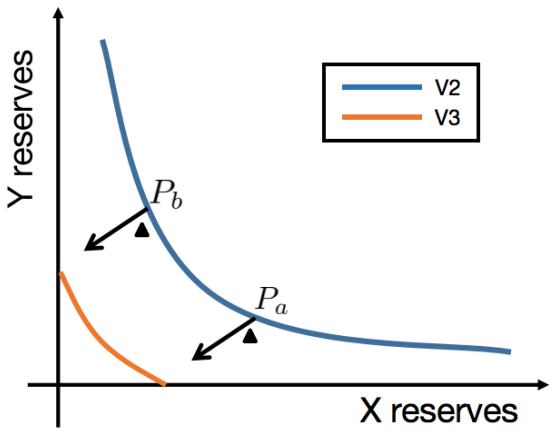
Uniswap governance

More Efficiency Protocols

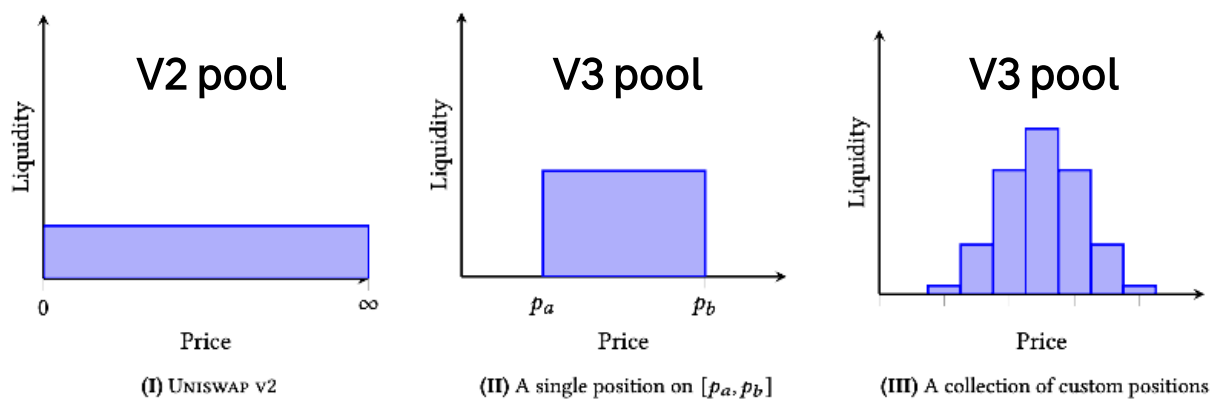
- This AMM mechanism is used in Uniswap **V2**.
- There are some of the latest swap protocols designed to improve liquidity efficiency:
  - Uniswap V3
  - Uniswap X

Uniswap V3 Protocol

- Compared to V2, liquidity providers (LPs) in the V3 protocol can select the price range for their liquidity.



- Significant improvement example: USDC – USDT pool



(I) Uniswap v2 (II) A single position on $[p_a, p_b]$ (III) A collection of custom positions

Uniswap V3 — Core Formulas

V2 Recap: Constant Product

$$x \cdot y = k$$

Liquidity spread across entire price range $[0, \infty)$

V3 Key Insight: Concentrated Liquidity

LP chooses price range $[P_a, P_b]$

Same capital → much higher virtual liquidity

Virtual Reserves Formula

$$\left(x + L / \sqrt{P_b} \right) \cdot \left(y + L \cdot \sqrt{P_a} \right) = L^2$$

Where: x, y = real token reserves | L = liquidity | P_a, P_b = lower/upper price bound

Real Reserves (what LP actually deposits)

$$x = L \cdot \left(1/\sqrt{P} - 1/\sqrt{P_b} \right)$$

$$y = L \cdot \left(\sqrt{P} - \sqrt{P_a} \right)$$

Uniswap V3: How is L Calculated?

When an LP deposits into a position on $[P_a, P_b]$, liquidity L is derived from the deposited amounts

Case 1: $P_a \le P \le P_b$

Price in range → deposit both X and Y

$$L_x = \Delta x / (1/\sqrt{P} - 1/\sqrt{P_b})$$

$$L_y = \Delta y / (\sqrt{P} - \sqrt{P_a})$$

$$L = \min(L_x, L_y)$$

Both tokens required.
Ratio set by P and range.

Case 2: $P < P_a$

Price below range → deposit only X

$$L = \Delta x / (1/\sqrt{P_a} - 1/\sqrt{P_b})$$

All X, no Y needed.
Position is 100% token X.

Case 3: $P > P_b$

Price above range → deposit only Y

$$L = \Delta y / (\sqrt{P_b} - \sqrt{P_a})$$

All Y, no X needed.
Position is 100% token Y.

Worked Example: Deposit 1 ETH into ETH/USDC pool, P = 2000, range [1500, 2500]

① Compute L from Δx = 1 ETH

$$L = 1 / (1/\sqrt{2000} - 1/\sqrt{2500})$$
$$= 1 / (0.02236 - 0.02000)$$
$$= 1 / 0.00236 \approx 423.7$$

② Required Δy from L

$$\Delta y = L \cdot (\sqrt{P} - \sqrt{P_a})$$
$$= 423.7 \times (44.72 - 38.73)$$
$$= 423.7 \times 5.99 \approx 2537.8 \text{ USDC}$$

③ Final Position

Deposit: 1 ETH + 2537.8 USDC

L ≈ 423.7

Narrower range → higher L
for the same capital

Example: V2 vs V3 Capital Efficiency

Scenario: ETH/USDC pool, current price P = 2000 USDC/ETH
LP provides liquidity in the range P_a = 1500 to P_b = 2500

	V2 (Full Range)	V3 [1500, 2500]
Capital Deposited	$10,000	$10,000
Liquidity (L)	$L_2 \approx 111.8$	$L_3 \approx 933.5$
Effective Liquidity Ratio	1×	~8.3×
Fee Earned (same volume)	$1	~$8.3

Calculation Walkthrough

① V2 Liquidity
$L_2 = \sqrt{(x \cdot y)} = \sqrt{(2.5 \times 5000)}$
≈ 111.8

② V3 Liquidity
L_3 = Capital / $(2\sqrt{P} - P/\sqrt{Pb} - \sqrt{Pa})$
≈ 933.5

③ Efficiency Gain
$L_3 / L_2 = 933.5 / 111.8$
≈ 8.3× more efficient!

Swap Example: 0.1 ETH → ? USDC

A trader swaps 0.1 ETH into the pool. Ideal output (zero slippage) = 0.1 × 2000 = 200 USDC. How close can each pool get?

V2 Pool (Full Range, L = 111.8)

Initial state:

$x = 2.5$ ETH, $y = 5000$ USDC

$k = x \cdot y = 12{,}500$

After swap (+0.1 ETH):

$x' = 2.5 + 0.1 = 2.6$

$y' = k / x' = 12500 / 2.6$

$= 4807.69$ USDC

Output:

$\Delta y = 5000 - 4807.69$

$= 192.31$ USDC

V3 Pool [1500, 2500], L = 933.5

Swap formula (√P form):

$\Delta x = L \cdot (1/\sqrt{P}_{new} - 1/\sqrt{P}_{old})$

$\Delta y = L \cdot (\sqrt{P}_{old} - \sqrt{P}_{new})$

Solve for √P_new:

$1/\sqrt{P}_{new} = \Delta x/L + 1/\sqrt{P}_{old}$

$= 0.1/933.5 + 1/44.721$

$\sqrt{P}_{new} = 44.508 \rightarrow P = 1980.9$

Output:

$\Delta y = 933.5 \times (44.721 - 44.508)$

$= 198.83$ USDC

Ideal: 200 | V2 output: 192.31 (slippage 3.85%) | V3 output: 198.83 (slippage 0.59%)

Larger Swaps: Slippage Comparison

As trade size increases, concentrated liquidity advantage becomes even more pronounced.

Swap Size (ETH)	Ideal Output	V2 Output	V2 Slippage	V3 Output	V3 Slippage
0.1 ETH	200 USDC	192.31	3.85%	198.83	0.59%
0.5 ETH	1,000 USDC	833.33	16.67%	977.7	2.23%
1.0 ETH	2,000 USDC	1,428.57	28.57%	1,909.4	4.53%

Key Takeaways

For Traders

V3 pools offer significantly better execution prices. Swapping 1 ETH: V3 returns 1,909 USDC vs V2's 1,429 USDC — saving $480 in slippage.

For LPs

Same $10,000 capital earns ~8× more fees in V3 because liquidity is concentrated where trades actually happen. Trade-off: if price leaves the range [P_a, P_b], the LP earns zero fees.

Uniswap V3 Protocol

■ Capital Efficiency Improvement

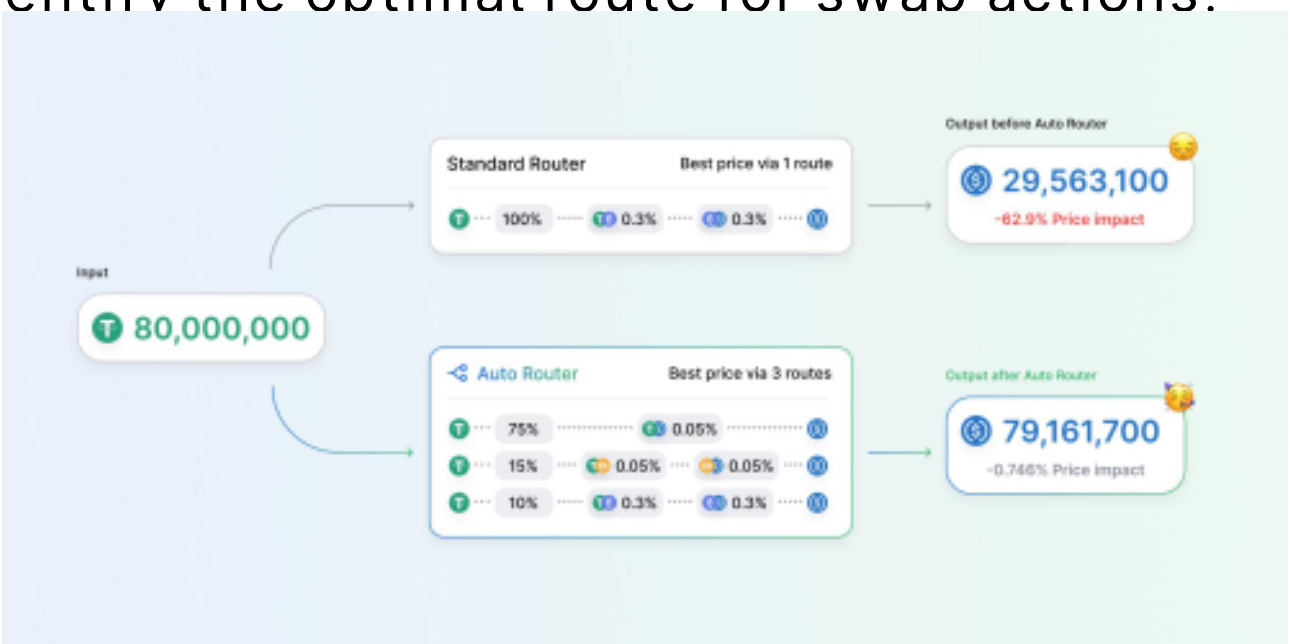
Compare to V2 range (0, +∞):



UniswapX Protocol

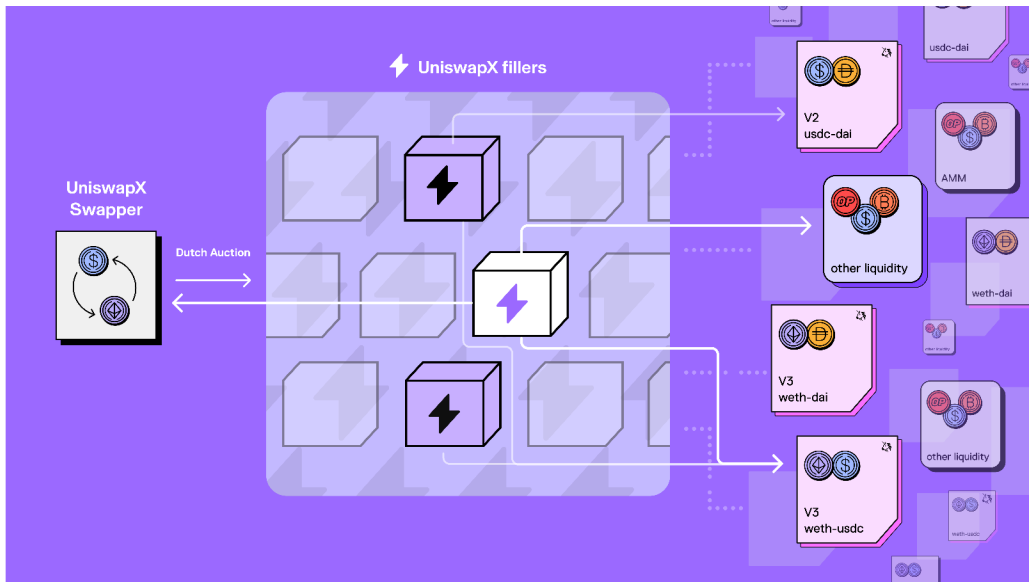
- Problem:

  Liquidity is fragmented across too many pools, making it difficult to identify the optimal route for swap actions.



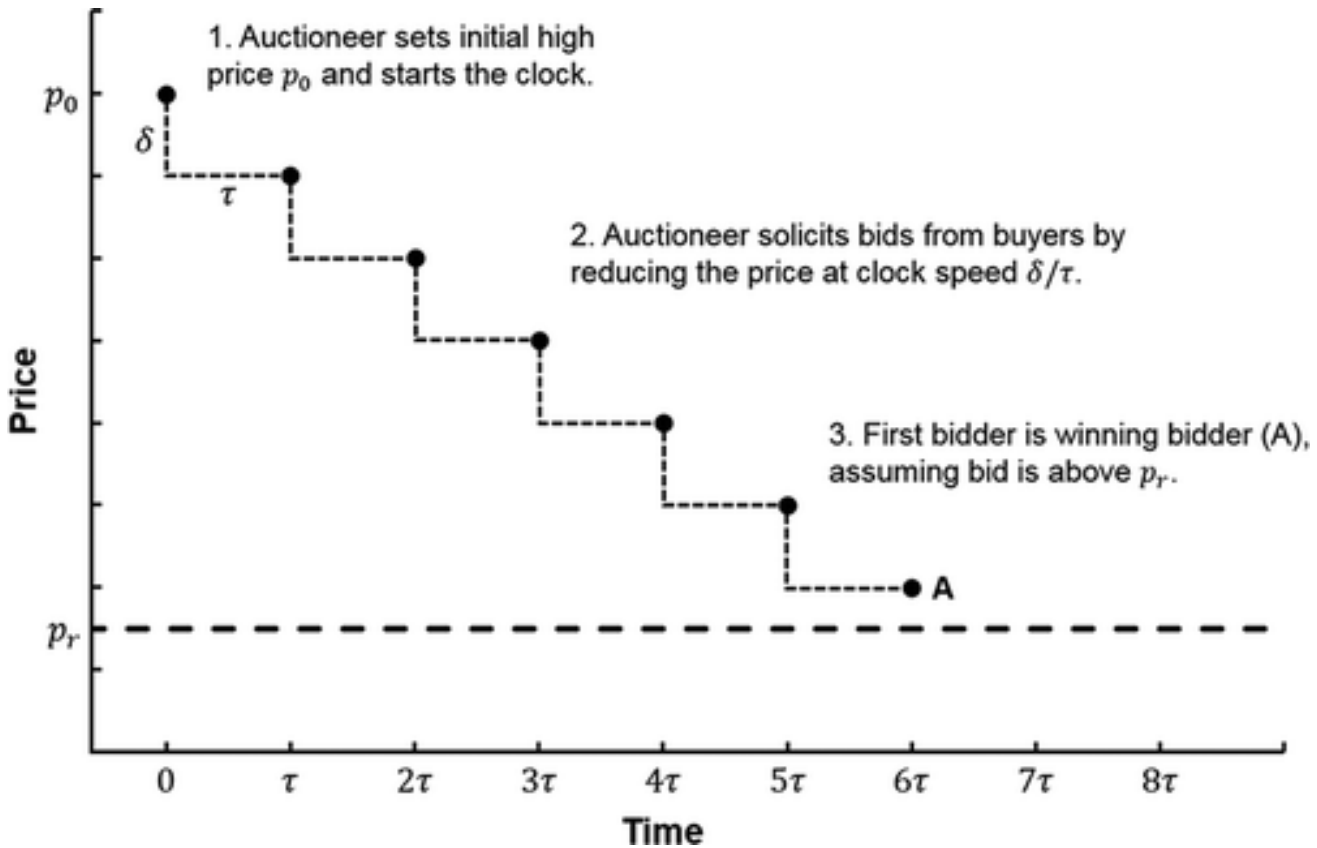
UniswapX Protocol

- UniswapX Protocol:
  - A permissionless, Dutch auction-based protocol designed for trading across AMMs and other liquidity sources.
  - A platform that aggregates liquidity sources to provide the best price for users.
  - Introduces third-party fillers to identify and provide the optimal route within the Dutch auction process.



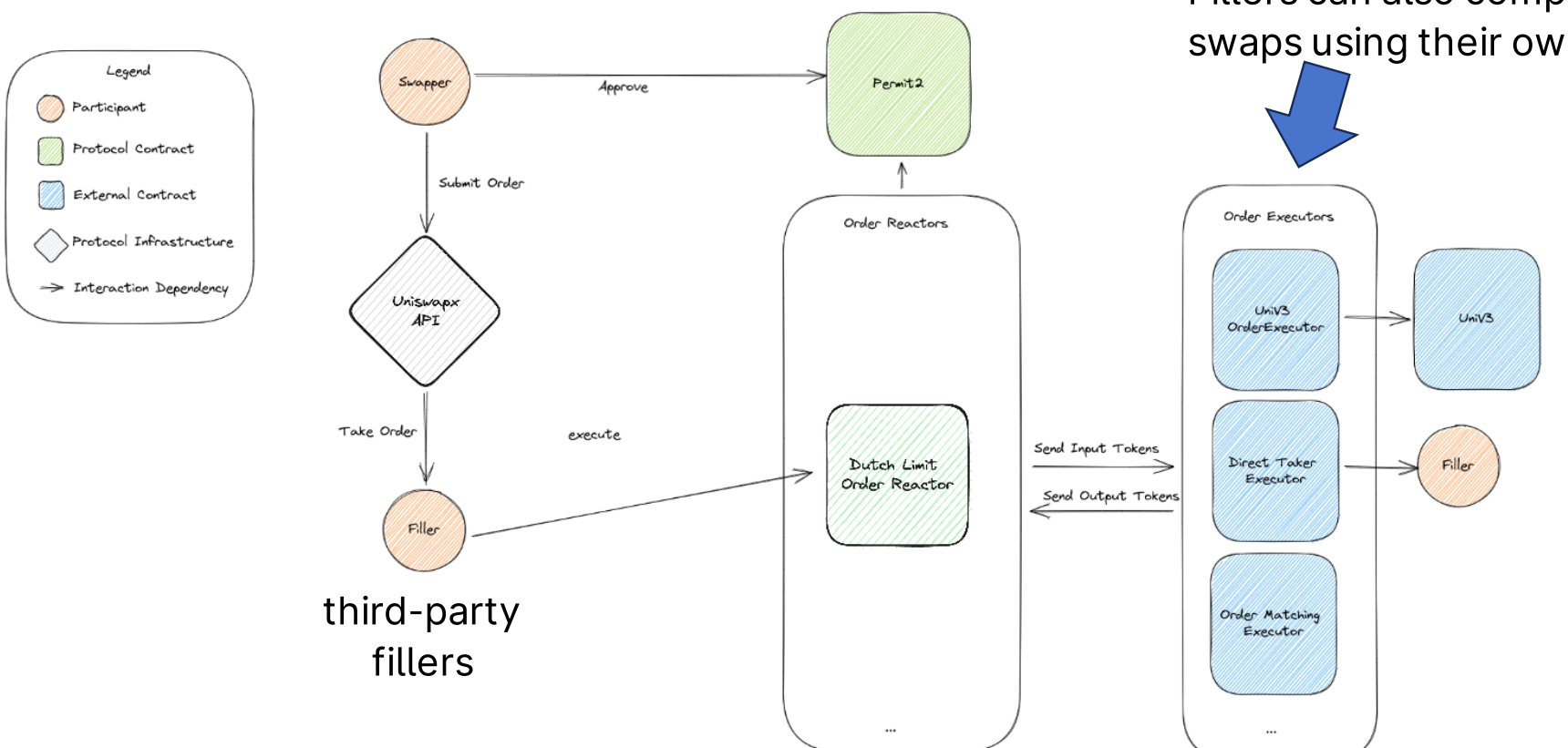
UniswapX Protocol

- Dutch Auction:



UniswapX Protocol

- UniswapX Architecture:



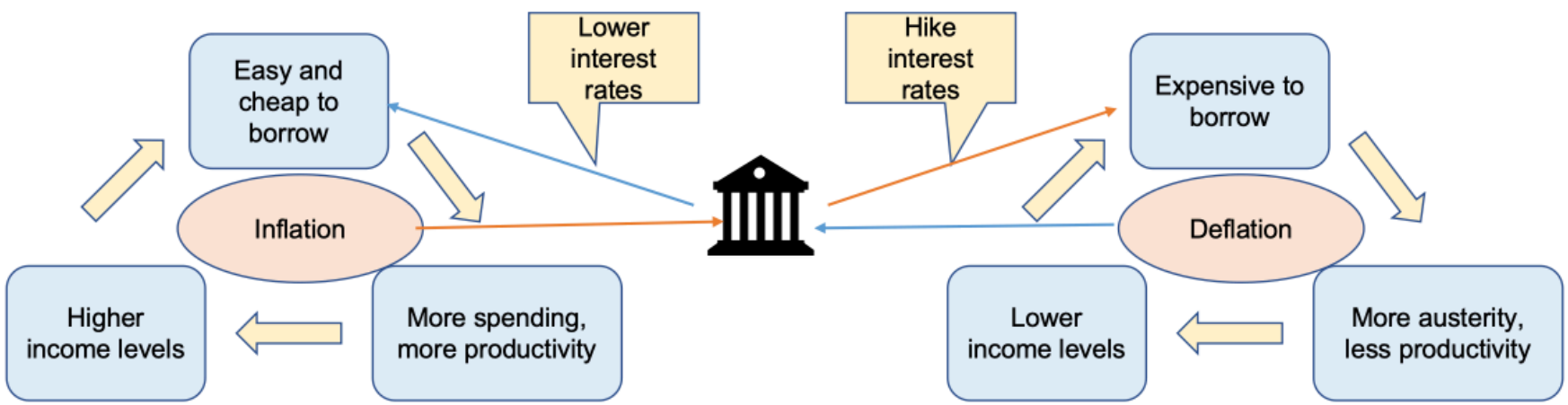
- All pools are accessible for use.
- Fillers can also complete swaps using their own assets.

Decentralized Finance



Need for lending

- Taking on debt enables borrowers to create value in future
- Created value is then used to pay back the debt
- Interest rates act as the “temperature control”



Lending in TradFi

- Banks lend out money
  - Trusted to lend out responsibly
  - Trusted to balance risk of defaults with enough assets/deposits

What if banks fail to do this?

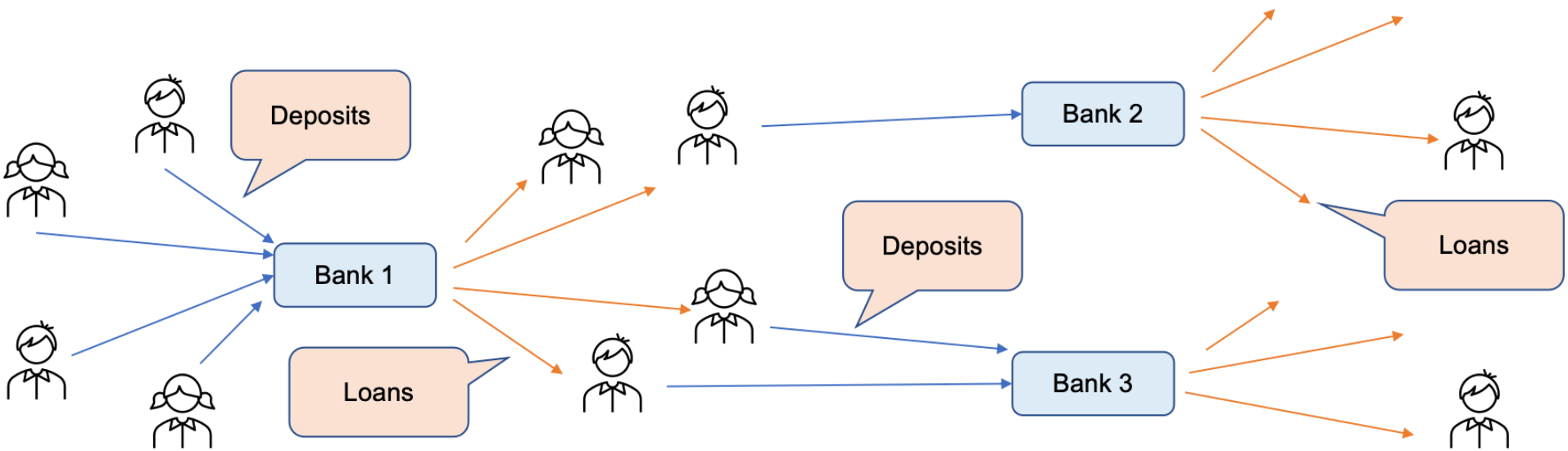
- Customers borrow money
  - Based on prior reputation or “credit score”
  - Based on income, assets

What if credit rating agencies are mistaken?

Lending in Tradfi: Monetary Multiplication

- Amount of active money in the economy gets multiplied by lending: $1 in the economy can be used to create $10 of productive value
- Leads to more capital efficiency
- But...

Defaults can lead to a chain reaction



Lending in TradFi: Vulnerability

2008 Financial Crisis: the perfect storm

- Banks did not lend out responsibly
- Credit rating agencies were mistaken about the credit scores
- Defaults did start a chain reaction

Need more transparency in

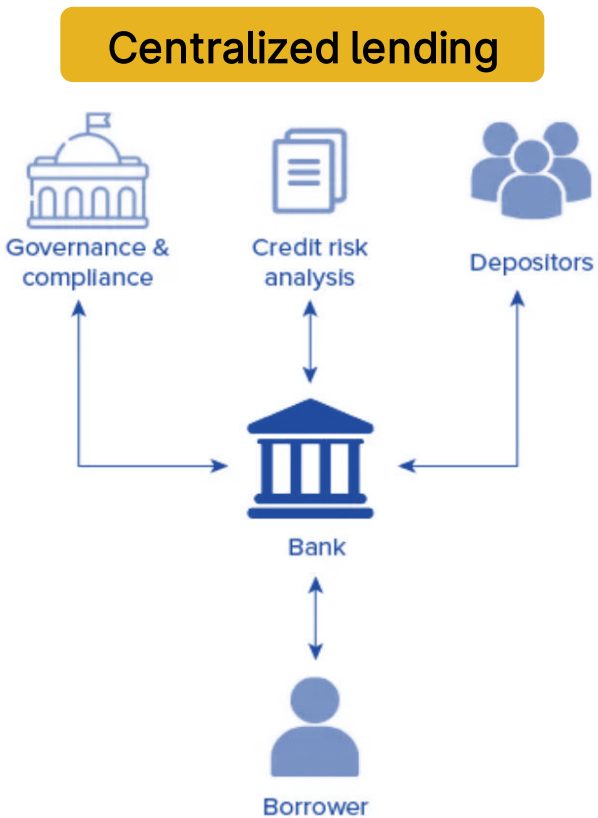
- The risks that lenders are taking on – deposits and collateral
- Metrics used to judge quality of loans – loan health factors

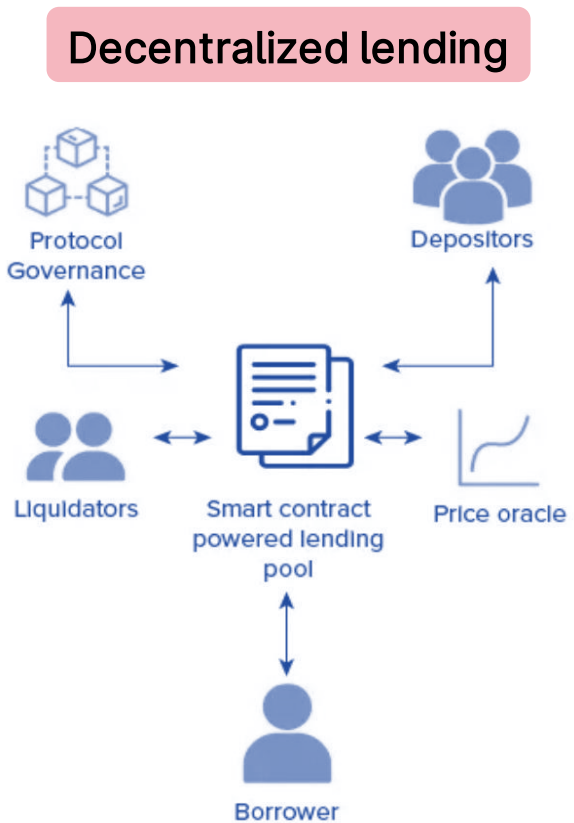
Decentralized Lending

- How can lending be enabled in a trustless + permissionless setting?
- Borrower required to lock in collateral in asset 2 for every asset 1 borrowed
- Protocol is on-chain and transparent – health of loans can be assessed easily
- Other agents incentivized to check if the assets and liabilities of a lending contract are balanced - liquidators

Decentralized Lending

Decentralized lending and borrowing markets allow any user to borrow or lend digital assets via decentralized protocols governed by smart contracts.





Decentralized lending: Setup

- Most DeFi lending currently is "over-collateralized"
  - Collateral
    - Assets that serve as a security deposit
  - Over-collateralization
    - Borrower has to provide value(collateral assets) > value(granted loan)

Why is Overcollateralization Needed in DeFi?

Example in real life:

Concern: what if Bob defaults on loan? ⟹ Bank will absorb the loss

Solution: require Bob to lock up collateral



Deposit a house worth over $250k

Borrow $100k



Bob

over collateralized loan

Why is Overcollateralization Needed in DeFi?

- By using smart contracts, DeFi lending protocols force borrowers to overcollateralize any loans they take out.
- Overcollateralization provides enhanced security for lenders in the DeFi space, significantly decreasing the risk of defaults due to the excess collateral that acts as a buffer during market downturns.

Collateral Factor

Collateral Factor$\in[0,1]$

- Max value that can be borrowed using this collateral

- High volatility asset $\Longrightarrow$ low collateral factor

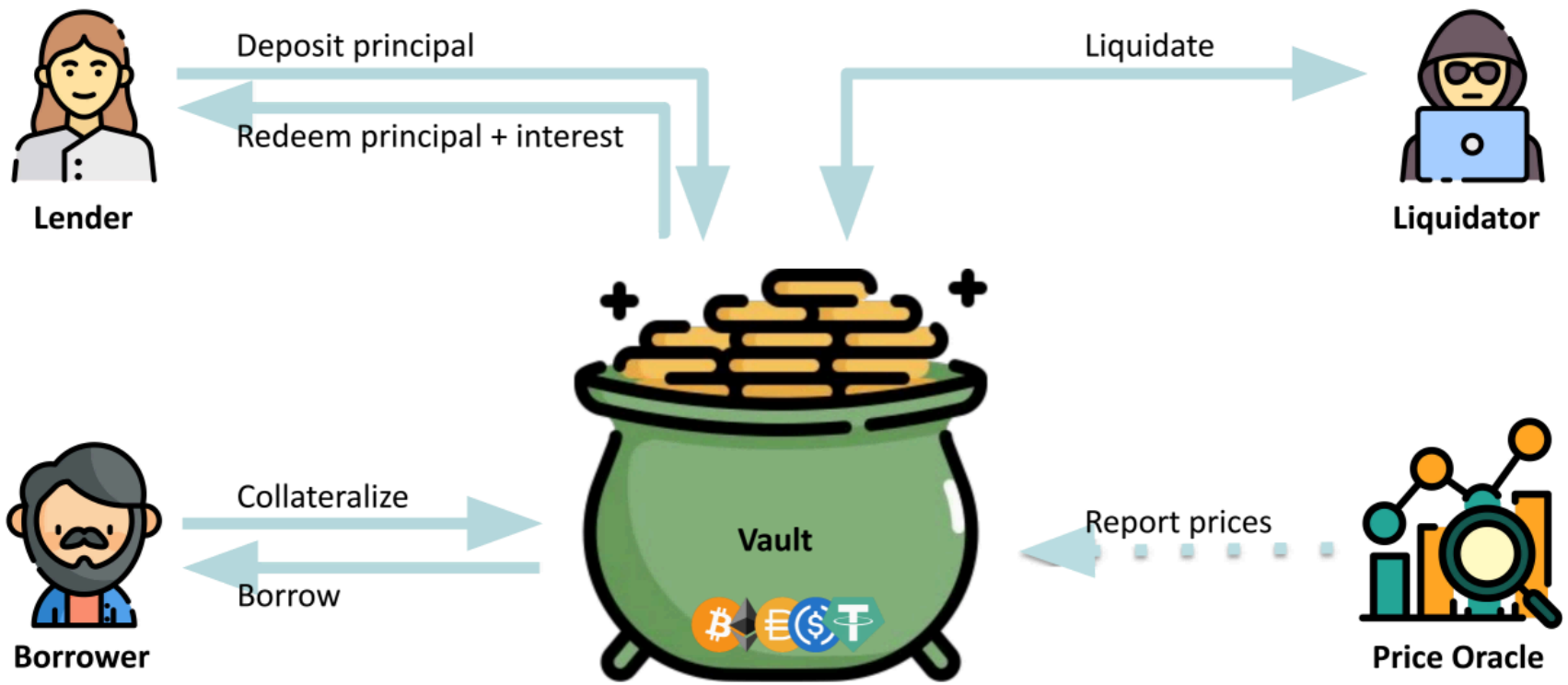
- Relatively stable asset $\Longrightarrow$ higher collateral factor

<u>Examples:</u> (on Compound)

ETH, DAI: 83%, UNI: 75%

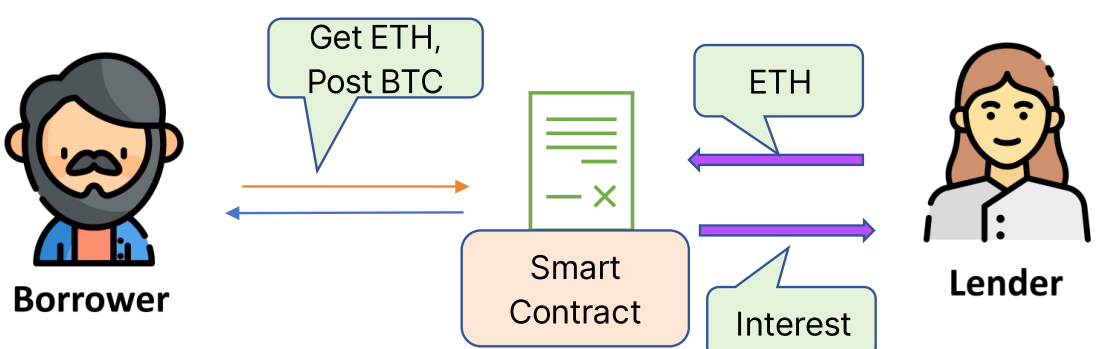
Agents Involved in Lending

Decentralized lending sets up incentives for the following agents that interact with each other to create a healthy lending environment.



Agents: Lender and Borrower

- Lender
  - Has some capital lying around and wants it to generate yield/interest
- Borrower
  - Has a need for capital that would generate value in the borrowed token
  - Incentivized to keep loan afloat with enough collateral, gets penalized otherwise



Lenders' Incentives: Interest Rates

- Each liquidity pool has utilization rate which is used to decide the interest rate

$$U_t = \frac{TotalBorrows}{TotalLiquidity}$$

Examples:

Pool: 1000 USDT ,Borrowed: 600 USDT $\rightarrow U_t = 60\%$

No borrowing $\rightarrow U_t = 0\ \%$

All funds borrowed $\rightarrow U_t = 100\%$

Lenders' Incentives: Interest Rates

- DeFi protocols use an interest rate model to adjust rates dynamically based on utilization.

Low Utilization Rate (Funds Abundant)

Scenario: Utilization rate is low. Interest rates drop.

Encourages borrowing and discourages new lenders (lower returns).

Example: Utilization = 20% → Interest Rate ≈ 2%.

Lenders' Incentives: Interest Rates

- DeFi protocols use an interest rate model to adjust rates dynamically based on utilization.

High Utilization Rate (Funds Tight)

Scenario: Utilization rate is high. Interest rates rise.

Attracts more lenders to add funds. Increases borrowing costs, limits over-borrowing.

Example: Utilization = 90% → Interest Rate ≈ 15% or higher.

Lenders' Incentives: Interest Rates

- Tradeoff: as liquidity becomes scarcer, lenders get higher rate of return, but might not be able to withdraw all of it in case of a bankrun
- Protocol tries to keep value of U_t near a fixed $U_{optimal}$ by controlling the interest rate R_t

Agent: Liquidator

- Part of the loan that is underwater is repaid by selling corresponding collateral at a discount – who does this?
- Liquidator
  - can liquidate the loan if value of collateral posted goes below threshold
  - needs to pay back part of the loan and gets corresponding part of collateral at a discount

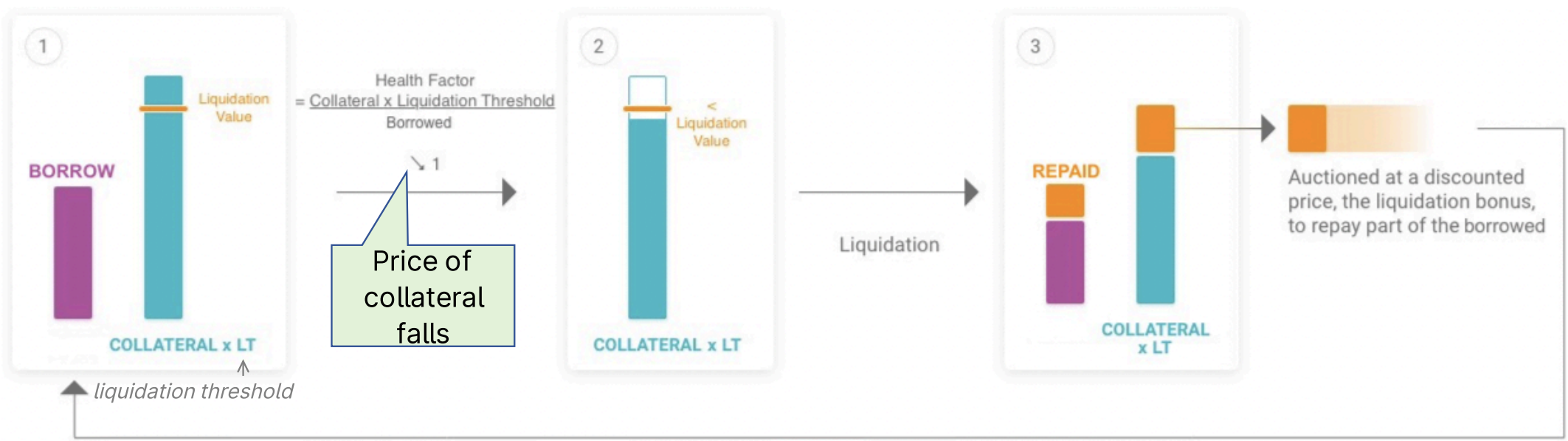
Agent: Liquidator

- Every token to be lent out has a liquidation threshold
  - It is preset by the protocol for each collateral asset (e.g., ETH = 85%).
  - It is an input value in the Health Factor calculation, defining the 'safety bottom line' for liquidation.
  - Health Factor in terms of collateral, threshold and borrowed amount:

$$H_f = C_{USD} \frac{L_{Threshold}}{B_{USD}}. \qquad \text{Example: } H_f = 100 \frac{0.75}{75} = 1$$

Agent: Liquidator

- Liquidation Threshold is a static rule, defining the boundary of 'when it will be liquidated'
- Health Factor is a dynamic tool for judging liquidation, directly telling you 'whether it will be liquidated now.
- If Health Factor H_f goes below 1, then liquidation (give the example later) is triggered

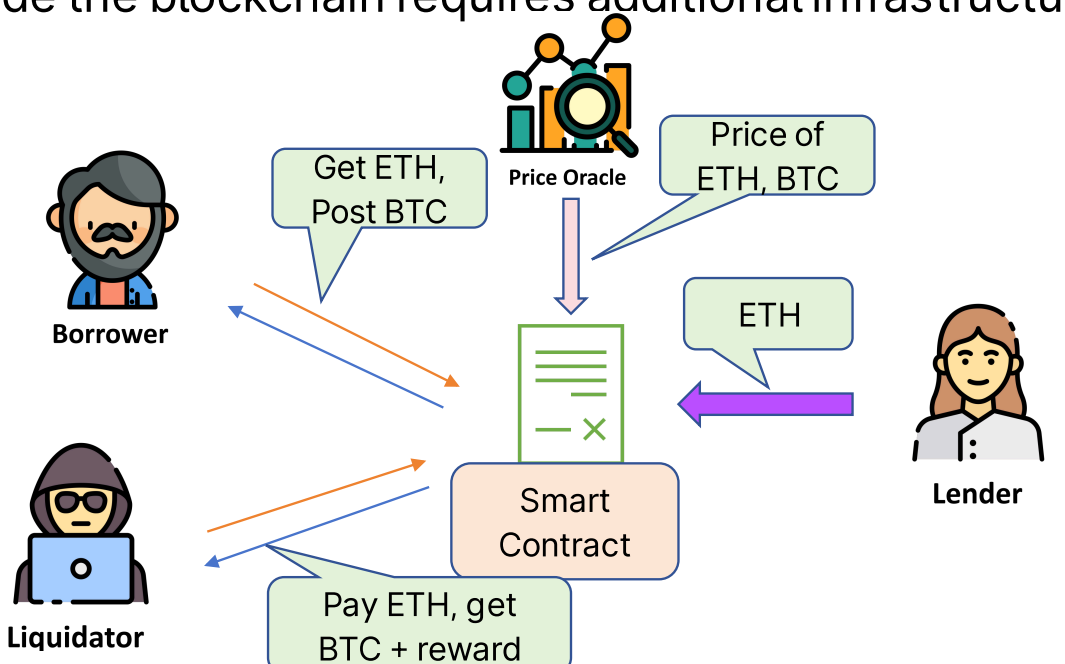


Agent: Oracle

How does smart contract know value of collateral vs loan?

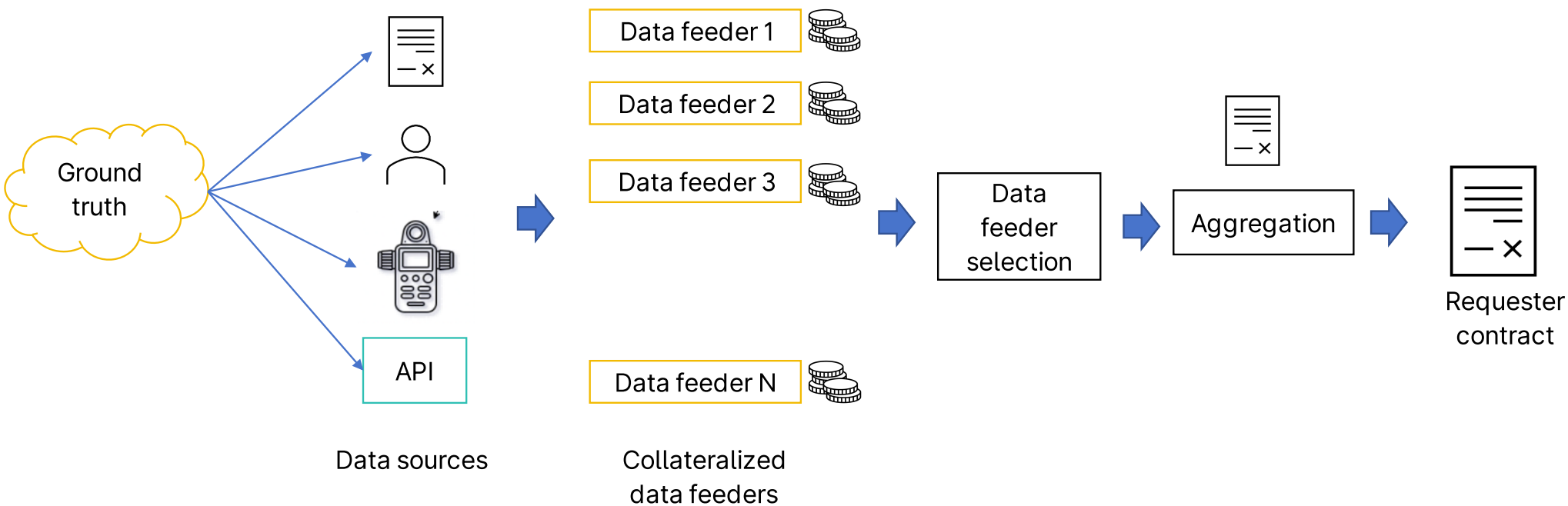
If liquidator wants to trigger liquidation, how does smart contract verify if value of the collateral is actually below threshold?

Accessing data outside the blockchain requires additional infrastructure: Oracle



Agent: Oracle

General Oracle Architecture



Agent: Oracle

Ground truth

- External data to be gathered by the oracle
- Ground truth should be visible to all participants in the oracle system
- Oracle is used if importing ground truth to the blockchain in a provable manner is either:
  - Impossible – Weather data
  - Expensive – State of another public blockchain

Agent: Oracle

Data source

- Measure and store ground truth
- Depending on the type of ground truth
  - Noise in measurements is tolerated – different weather monitoring stations
  - Noise may not be tolerated – winner of a publicized game, state of a finalized ledger
- Examples: Sensors, humans, other smart contracts, databases

Agent: Oracle

Data feeder

- Reports off-chain data to on-chain oracle contracts
- Incentivized by collateralization with staking rewards proportional to collateral or reputation
- Collateralization prevents sybil attacks
- Data feeder reporting may be periodic or triggered by an on-chain request

Agent: Oracle

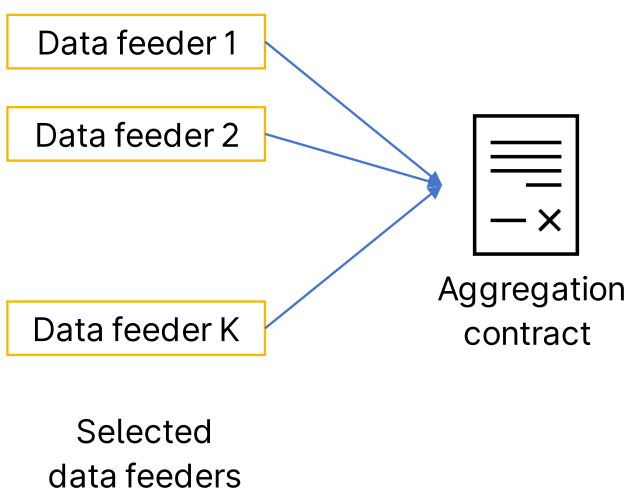
Data feeder selection

- The process of selecting what data-feeders report
- Selection needed for reputation incentivization and reducing on-chain transaction load
- Centralized selection: Restricts to a selected set of data feeders
- Decentralized selection: Random selection proportional to collateral/reputation

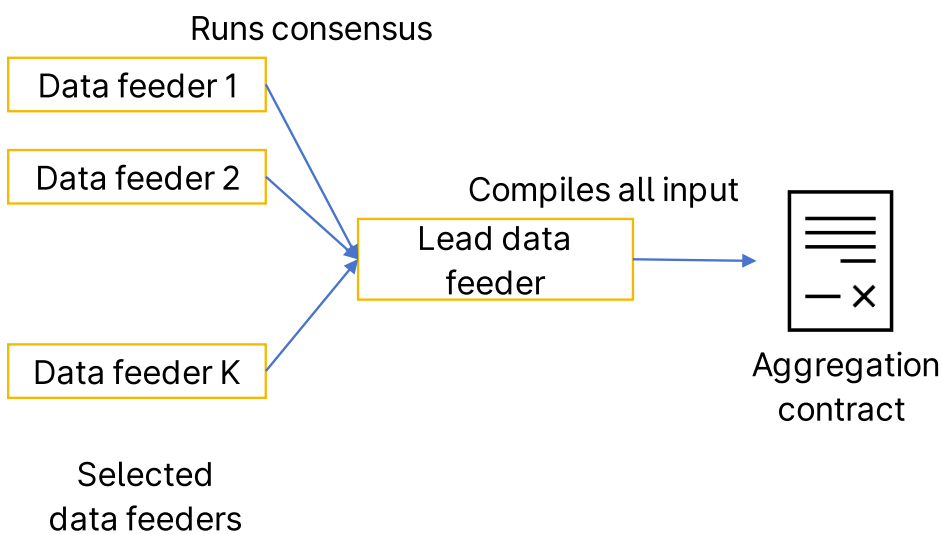
Agent: Oracle

Aggregation

- Oracle contract receives a lot of reports from data feeders
- All inputs need to be aggregated to be used by a DeFi contract
- Two types of input reporting:



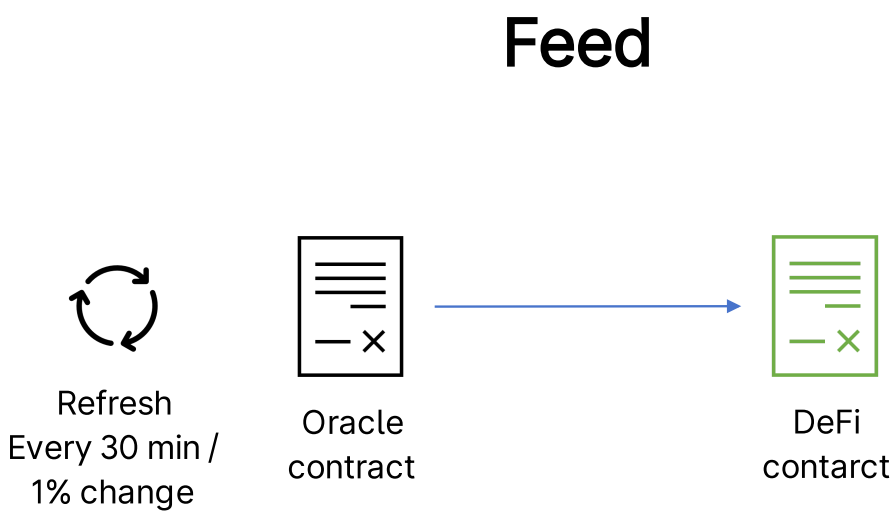
Direct reporting



Off-chain reporting

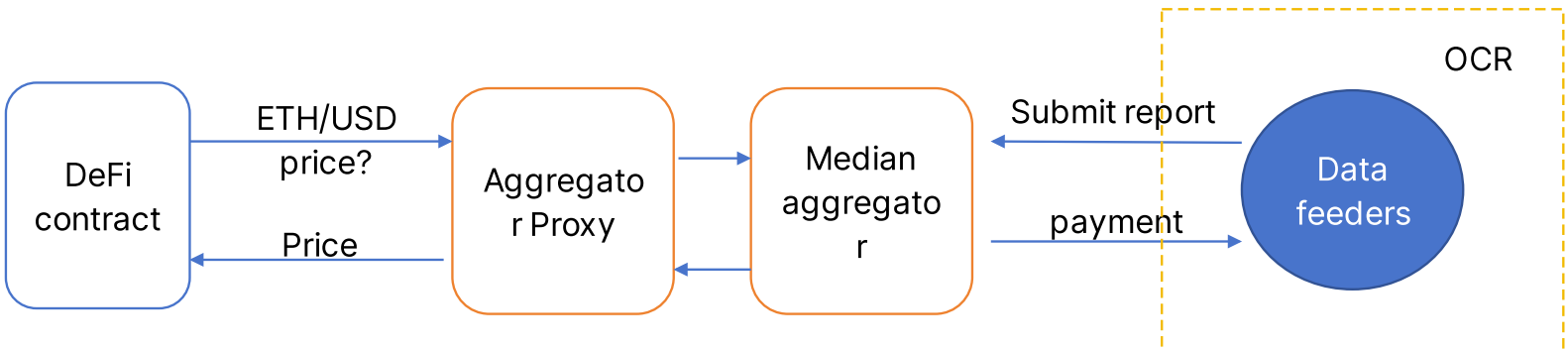
Agent: Oracle

Oracle interaction models



Agent: Oracle

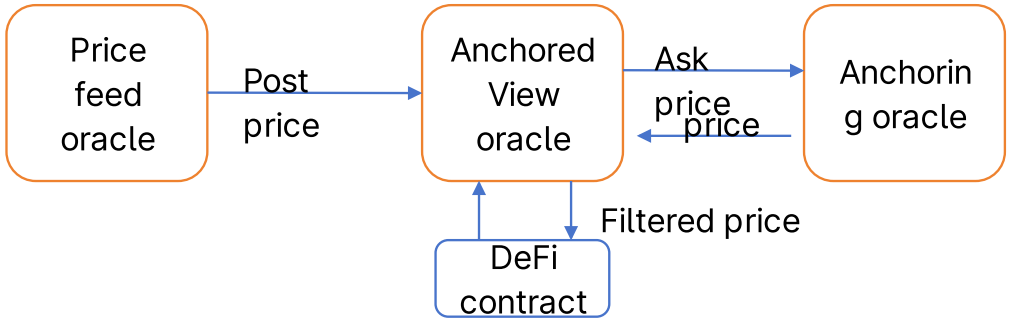
ChainLink price feed oracle design



- Reports are submitted at fixed intervals or when price deviates a threshold percentage from last reported price
- Data feeders maintain a consensus amongst themselves and the leader reports data to the aggregator (off-chain reporting)

Agent: Oracle

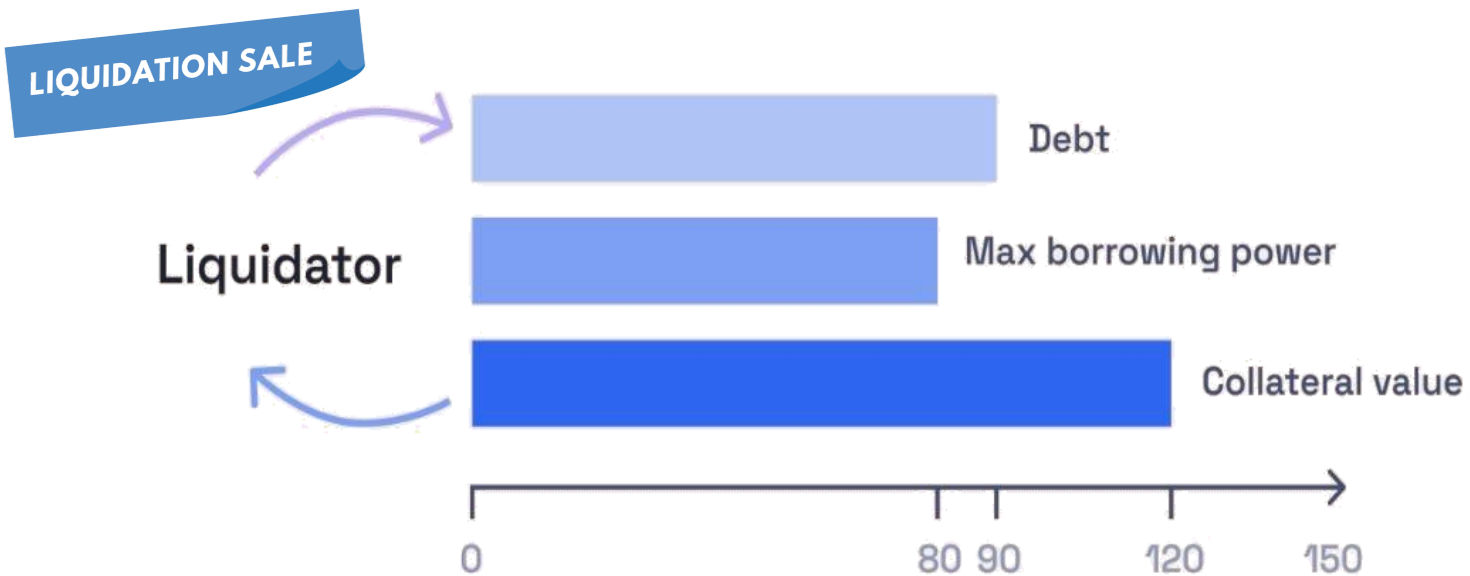
Compound oracle design



- If (Price feed posted price close to anchoring oracle price):
  - Filtered price = Price feed posted price
- else:
  - Filtered price = anchoring oracle price
- Anchoring oracle is typically an on-chain AMM (protection against price

Action Space: Liquidation

Liquidation

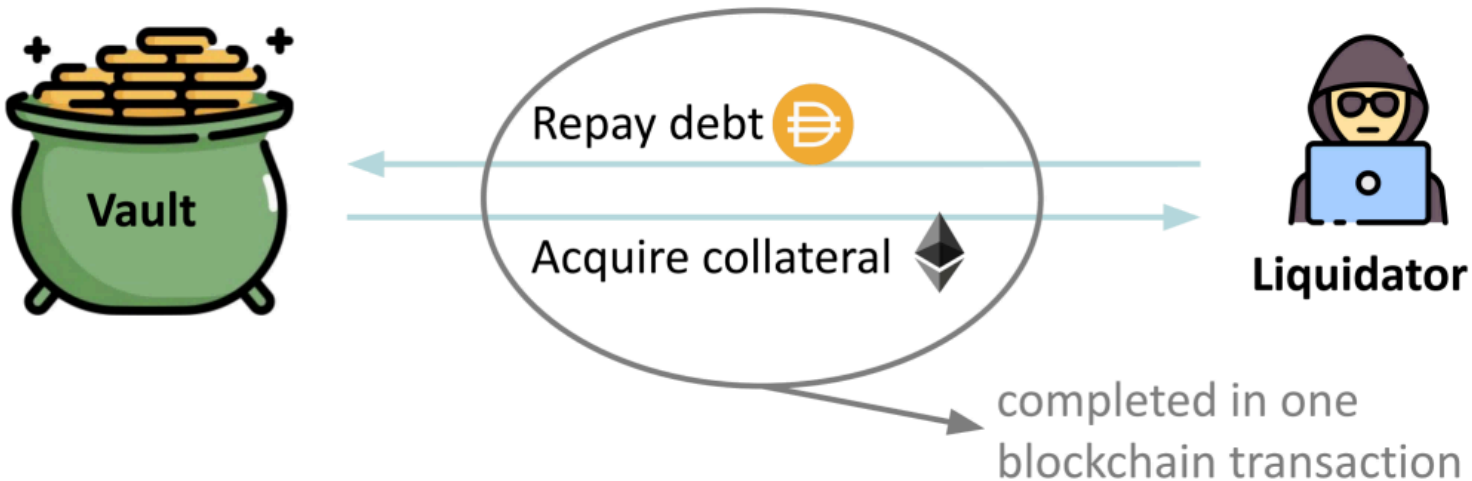


- The debt can get liquidated if the value of the collateral deposited gets too low compared to the value of the debt borrowed.
- Liquidation is the mechanism by which external actors repay the debt of borrowers on their behalf when they become insolvent.

Liquidation

- Liquidation == Selling collateral from the borrower
- Liquidation Spread
  - Bonus, or discount for liquidator
  - Fixed spread, or variable (auction based)

Fixed Spread Liquidation



- Repays the debts of a borrowing position
- Acquires the collateral at a discounted price from the position in return
- Typical discounts are e.g.,5-15%in Aave

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1000 DAI
Actual CDP	500 DAI
Collateral ratio	340%



Oracle

(2025.01)
1 ETH= 1700 DAI

Collateral ratio (CR) = $\frac{\text{Collateral value}}{\textit{Actual CDP}}$

If CR < LR, Alice's loan will be liquidated

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1000 DAI
Actual CDP	500 DAI
Collateral ratio	340%



Oracle

(2025.01)
1 ETH= 1700 DAI



(2025.02)
1 ETH = 800 DAI

Collateral ratio (CR) = $\frac{\text{Collateral value}}{\textit{Actual CDP}}$

If CR < LR, Alice's loan will be liquidated

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.02
Collateral	1 ETH
Collateral value	800 DAI
Maximum CDP	470.59 DAI
Actual CDP	500 DAI
Collateral ratio	160% < 170%



Oracle

(2025.02)
1 ETH = 800 DAI



Liquidation

If CR < LR, Alice's loan will be liquidated

Collateral ratio (CR) = $\frac{\text{Collateral value}}{\textit{Actual CDP}}$ = 800/500 = 160%

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.02
Collateral	1 ETH
Collateral value	800 DAI
Maximum CDP	470.59 DAI
Actual CDP	500 DAI
Collateral ratio	160% < 170%

Repays the loan

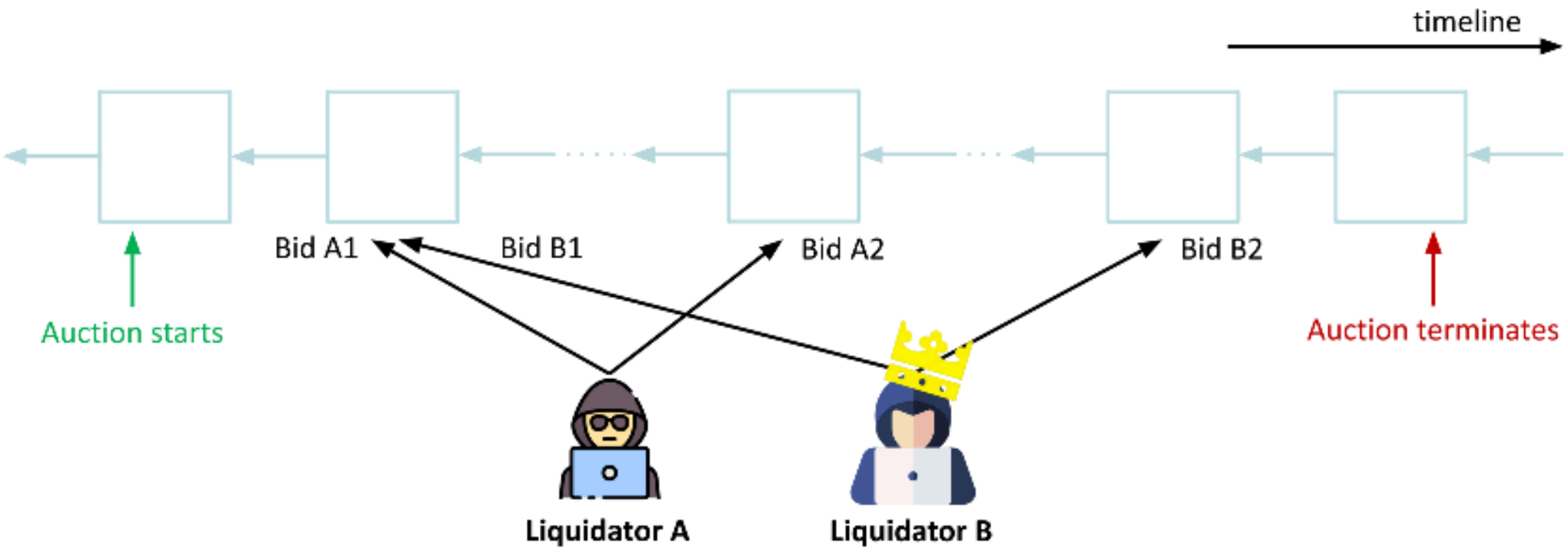


Liquidator

Liquidator could use 800*(1-5%) = **760 DAI** to buy **1 ETH (worth 800 DAI)**, and immediately sells this 1 ETH at the market price of 800 DAI with 0.001 ETH gas fee.

Auction Liquidation

- Various liquidators bid over time until the auction terminates
- Requires multiple blockchain transactions



Auction Liquidation

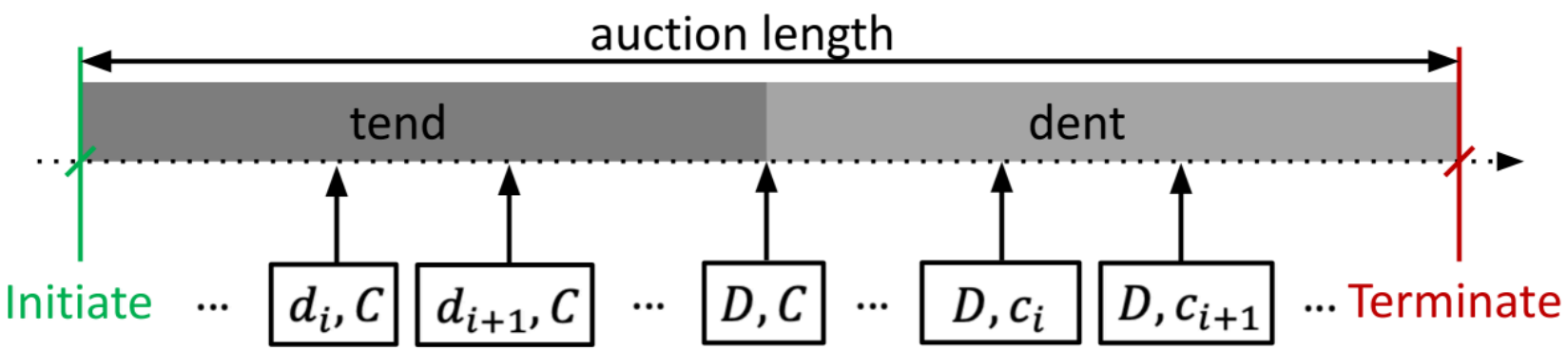
- **English Auction**
- bidders outbid each other increasingly

- **Dutch Auction**
- auction begins with a high asking price and the price lowers until the auction terminates

Auction Liquidation

MakerDAO tend-dent English auction

- A position with D debt and C collateral

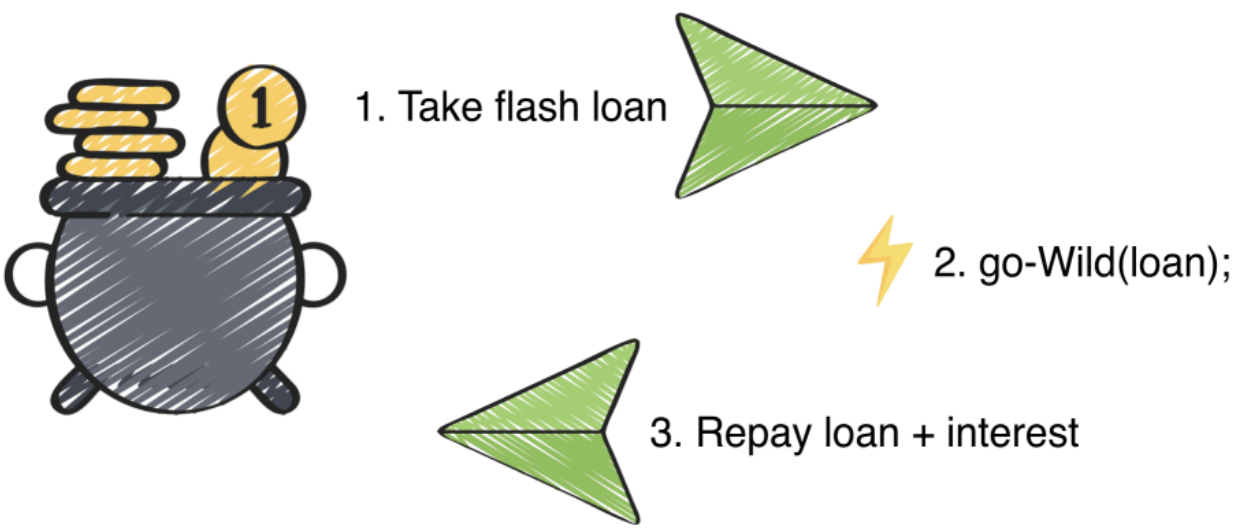


Flash loans: lending without collaterals

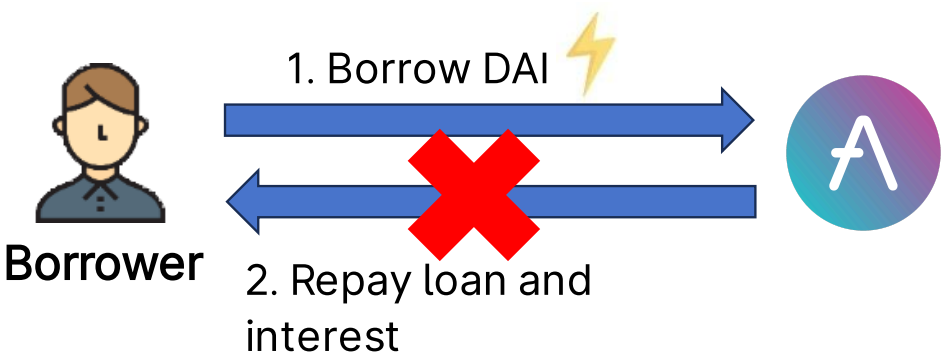
Flash Loans

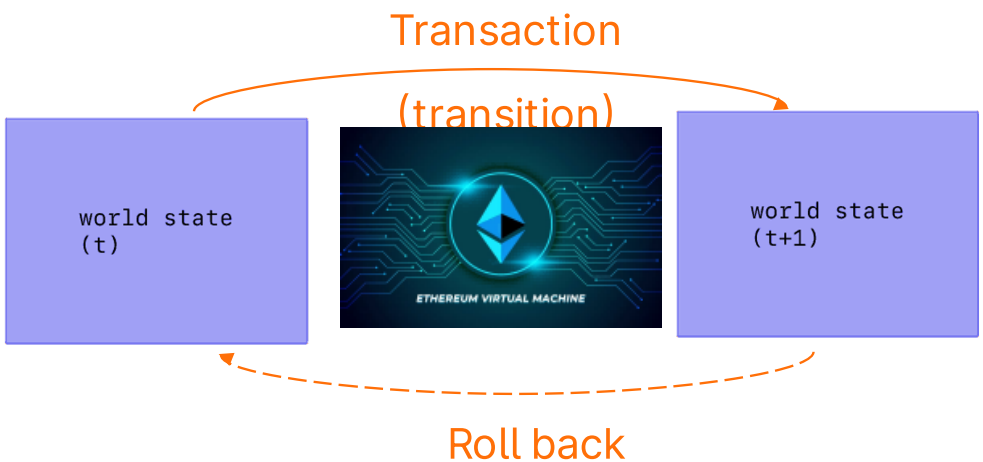
Flash loans are a special form of lending. A flash loan should be taken and repaid in one single transaction.

- Smart contract enable **atomic** transactions
- The actions within a transaction are executed entirely in sequence or fail collectively.

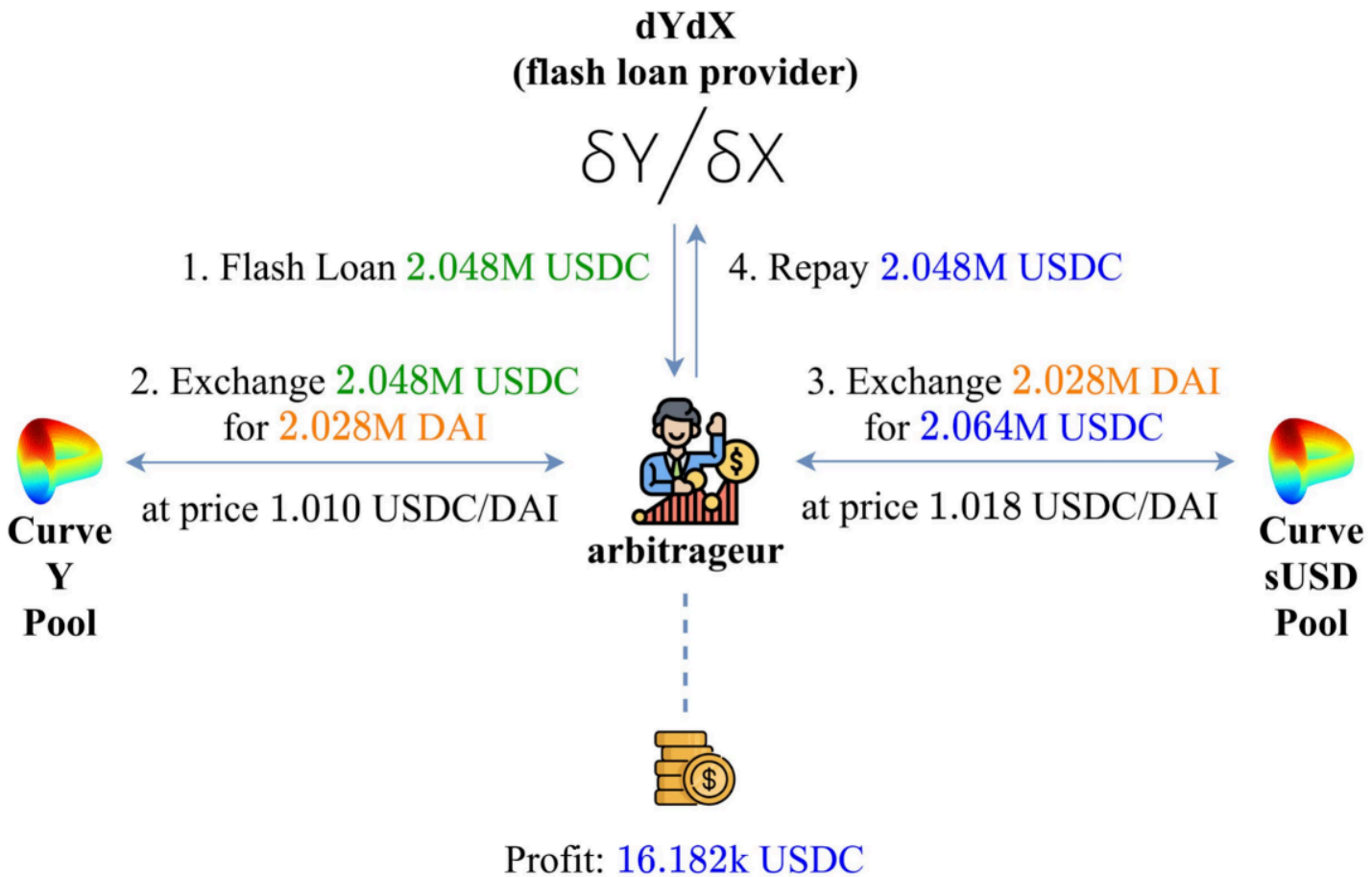


Flash Loan Example





Flash Loan Use Cases - Arbitrage



Flash Loan Use Cases - Flash Loan Based Liquidation

- When a liquidator does not have the cryptocurrency upfront to repay
- Only works when the liquidation completes in one transaction

Given a liquidatable borrowing position with a debt of 1000 DAl collateralized by 1 ETH

Flash Loan Use Cases - Flash Loan Based Liquidation

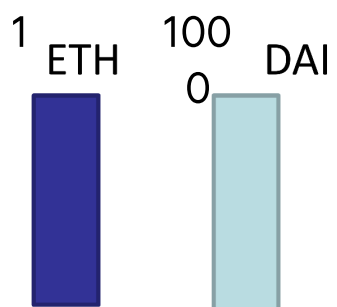
Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 130%

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1300 DAI
Actual CDP	1000 DAI
Collateral ratio	170%>130%

Collateral ratio (CR) = $\frac{\text{Collateral value}}{\textit{Actual CDP}}$

If CR < LR - liquidation



1 ETH= 1700 DAI

Flash Loan Use Cases - Flash Loan Based Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{\textit{Maxmium Collateralized Debt Position (CDP)}}$ = 130%

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1300 DAI
Actual CDP	1000 DAI
Collateral ratio	170%

1 ETH= 1200 DAI

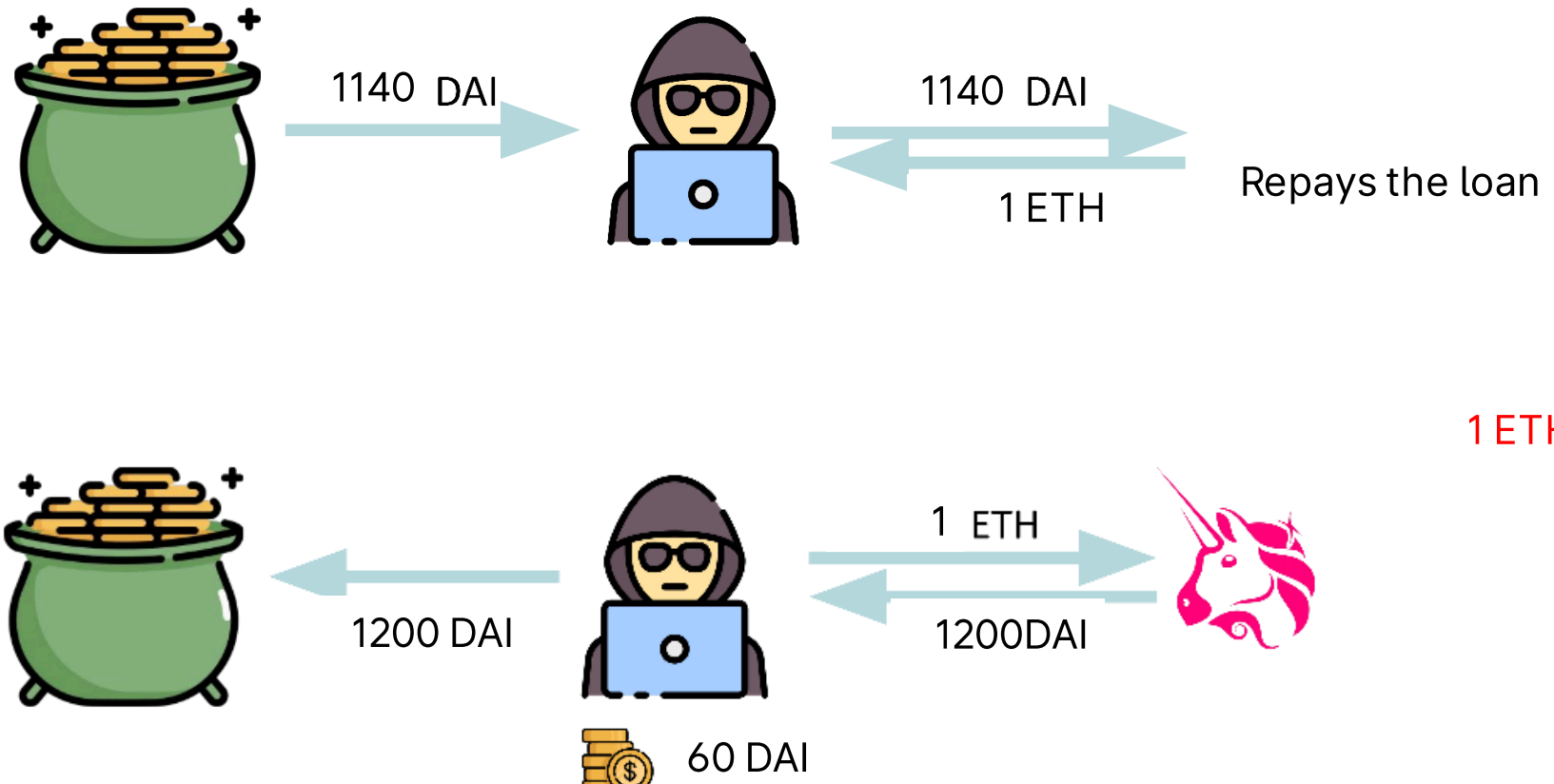
Date	2025.02
Collateral	1 ETH
Collateral value	1200 DAI
Maximum CDP	923.08 DAI
Actual CDP	1000 DAI
Collateral ratio	120% < 130%

Collateral ratio (CR) = $\frac{\text{Collateral value}}{\textit{Actual CDP}}$

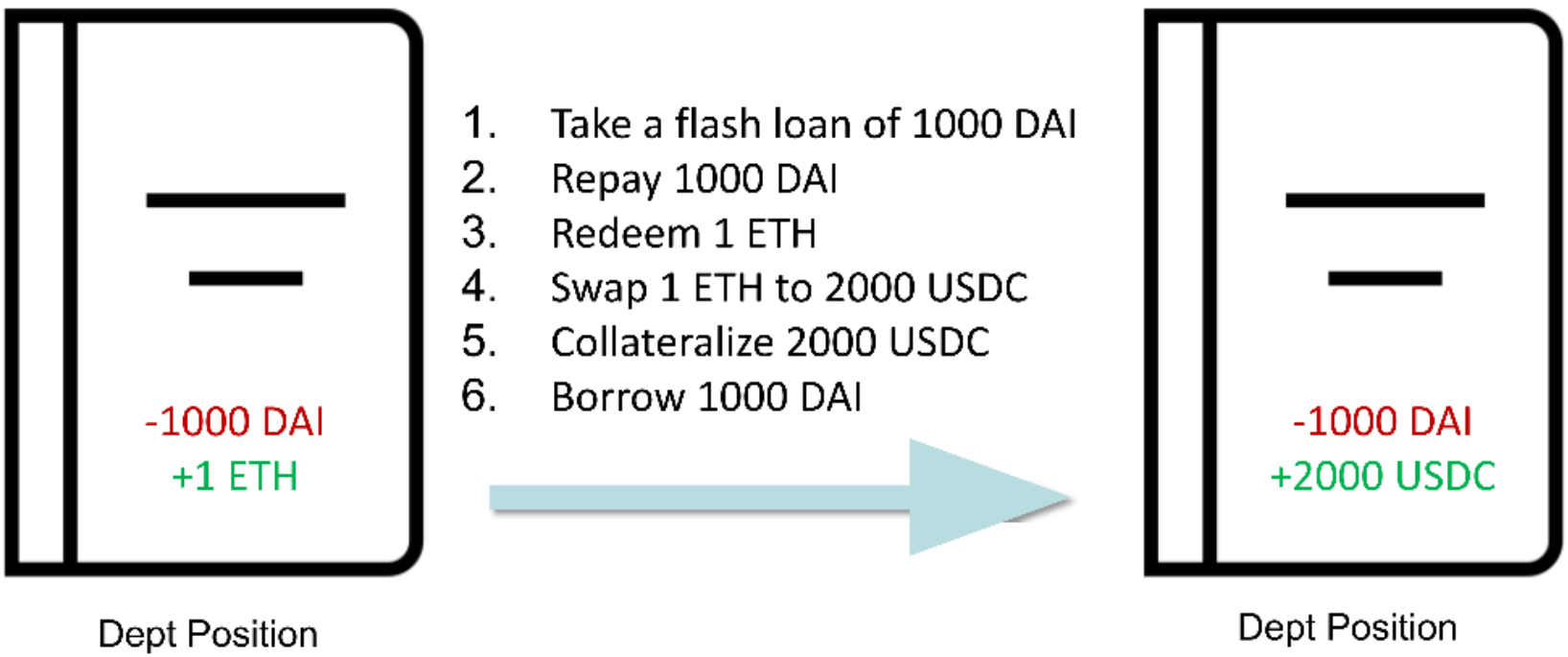
If CR < LR - liquidation

Liquidator could use 1200*(1-5%) = **1140 DAI** to buy **1 ETH**

Flash Loan Use Cases - Flash Loan Based Liquidation



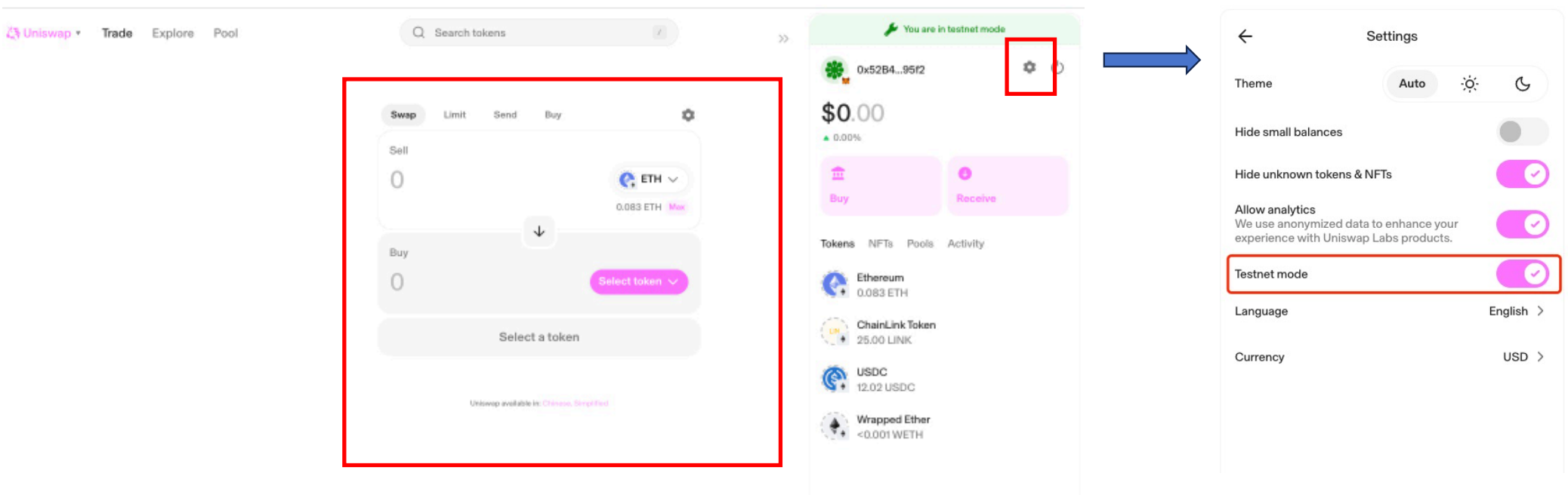
Flash Loan Use Cases - Collateral Swap



Workshop

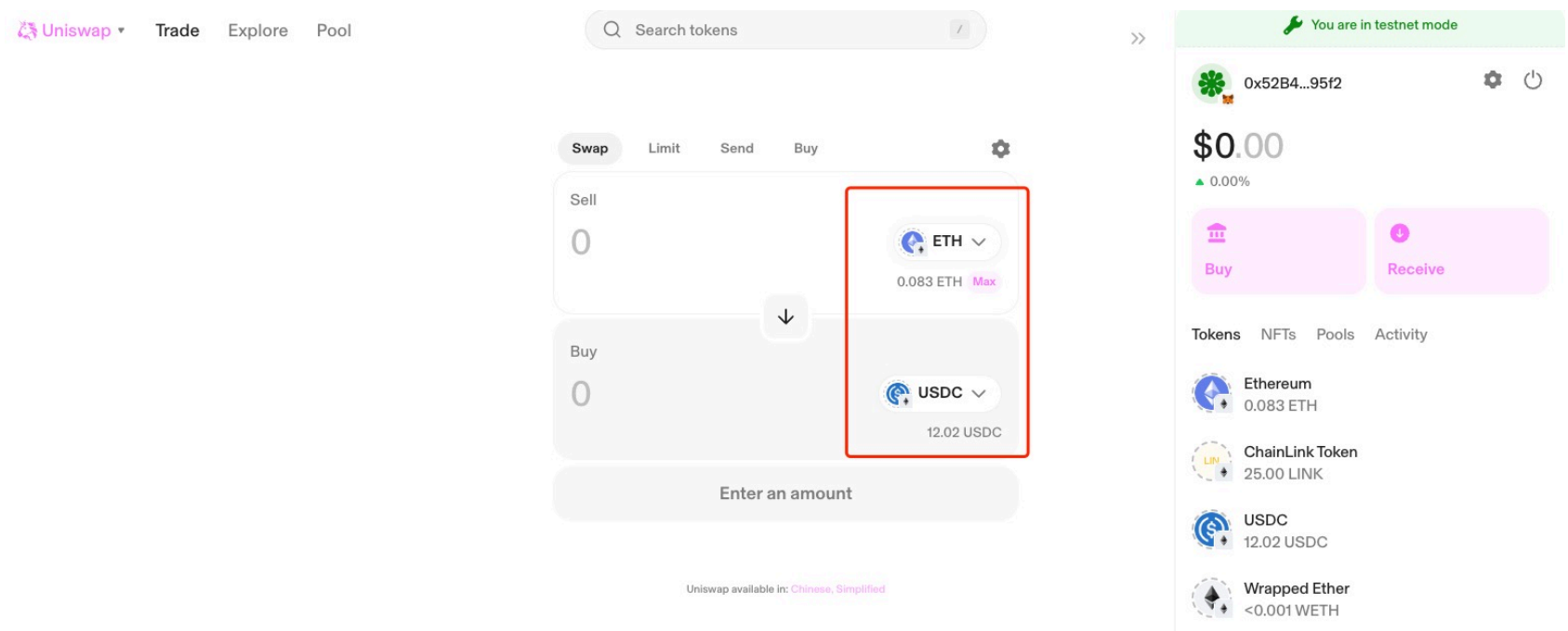
Interaction with Uniswap

- **Swap**: allows users to exchange one cryptocurrency for another directly, without relying on a centralized exchange like Binance or Coinbase.
  - Precondition: create a Web3 wallet and use testnet mode.



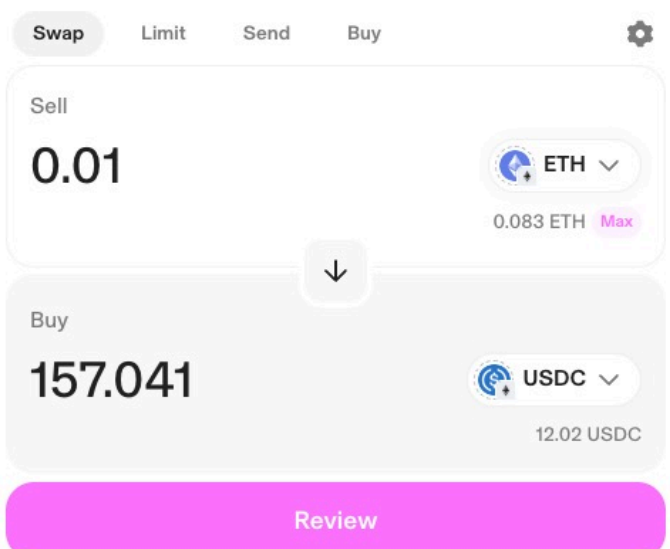
Interaction with Uniswap

- On the “Swap” page, select the token pair you want to exchange (e.g., ETH → USDC).



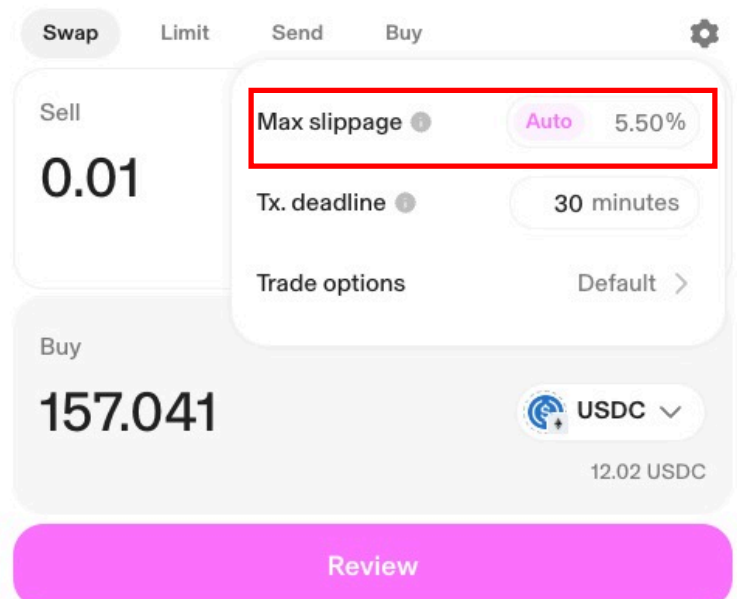
Interaction with Uniswap

- Enter the number of ETH to be exchanged, and the system automatically calculates the number of available USDC.



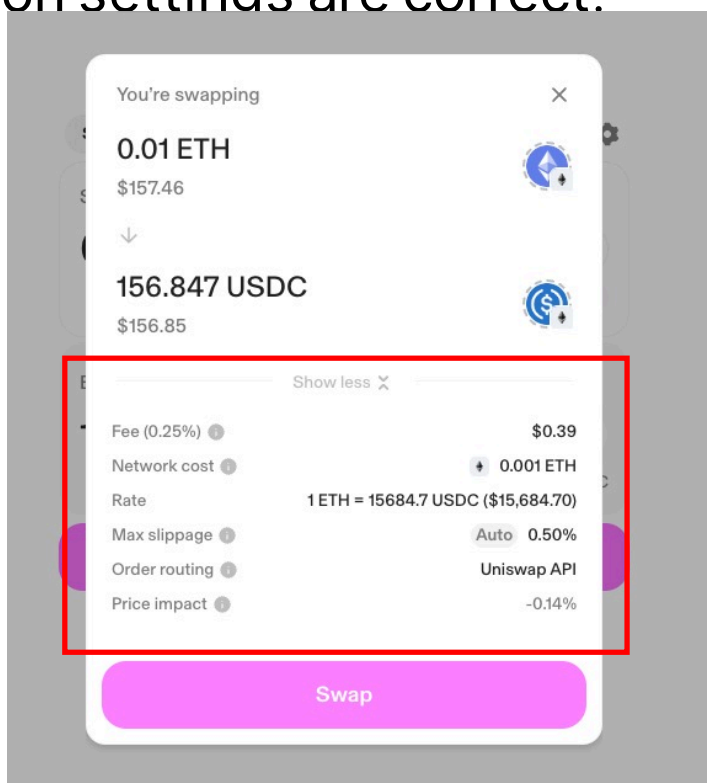
Interaction with Uniswap

- Check transaction details.
- Slippage: refers to the difference between the expected price of a trade and the price at which the trade is executed.



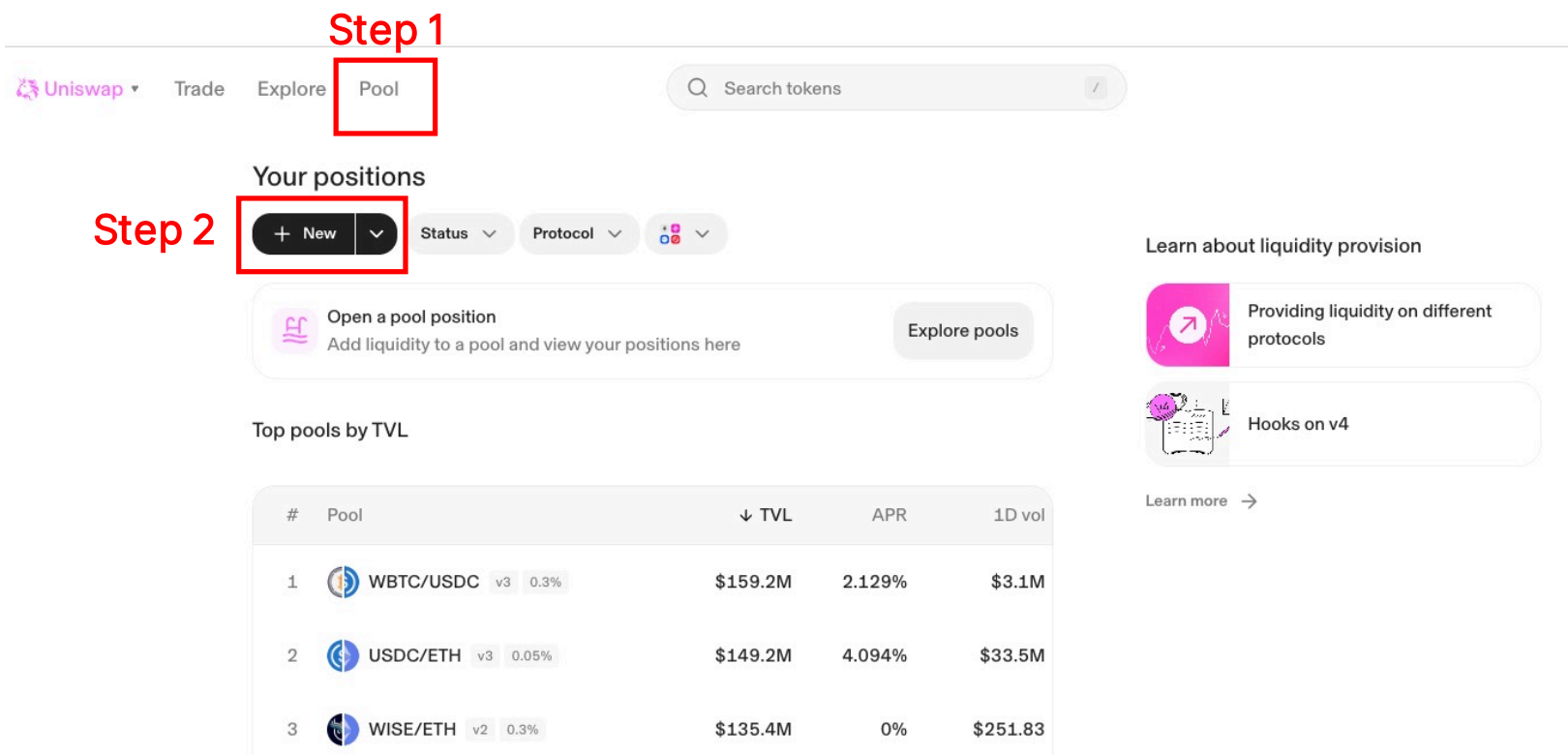
Interaction with Uniswap

- Review the transaction and remember to click “Show more” to double check that the transaction settings are correct.



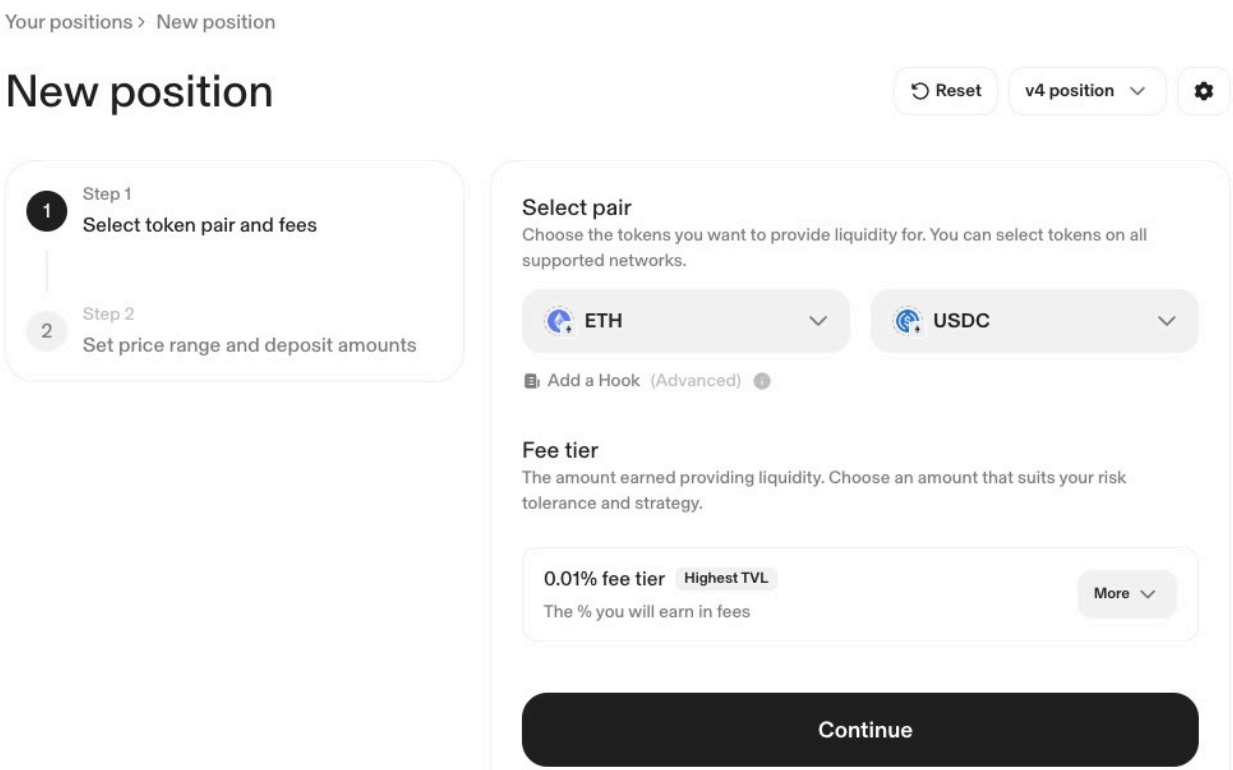
Interaction with Uniswap

- **Liquidity Providing:** Go to the “Pool” page and click on “New Position”.



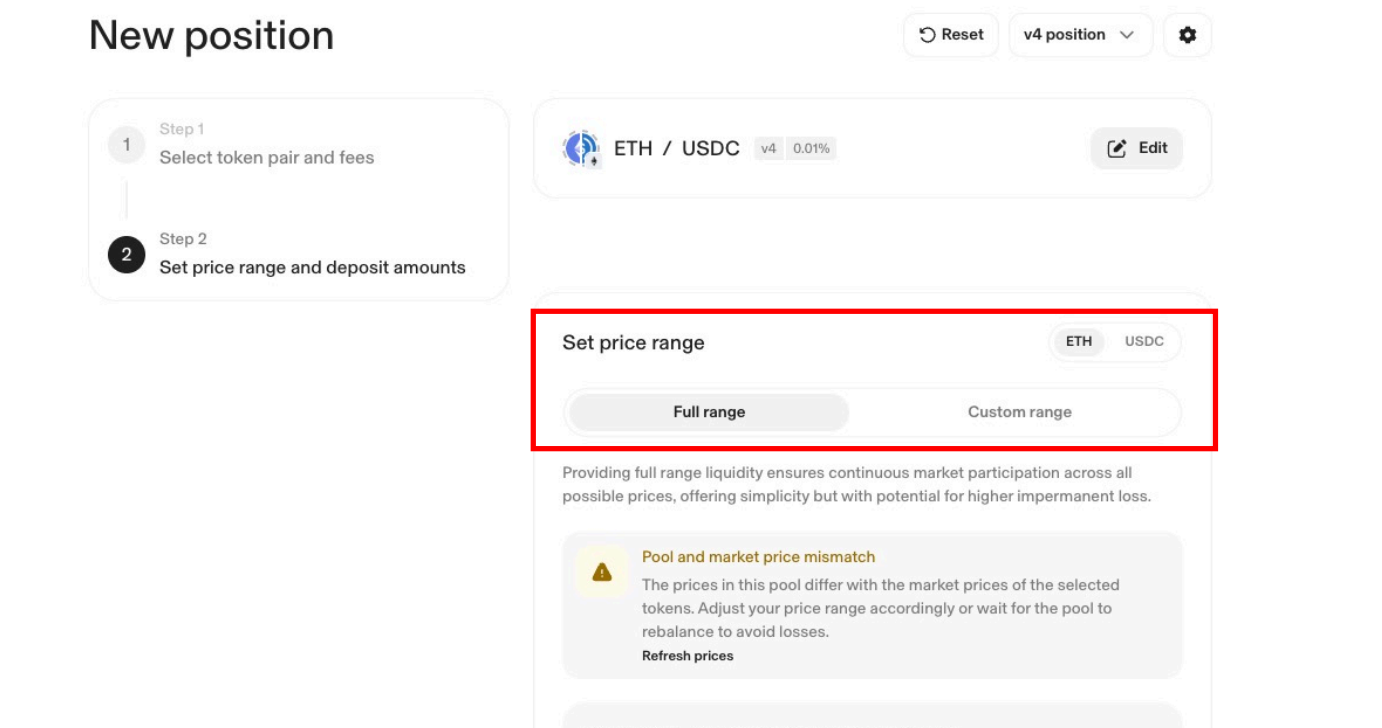
Interaction with Uniswap

- Select a token pair (such as ETH-USDC).



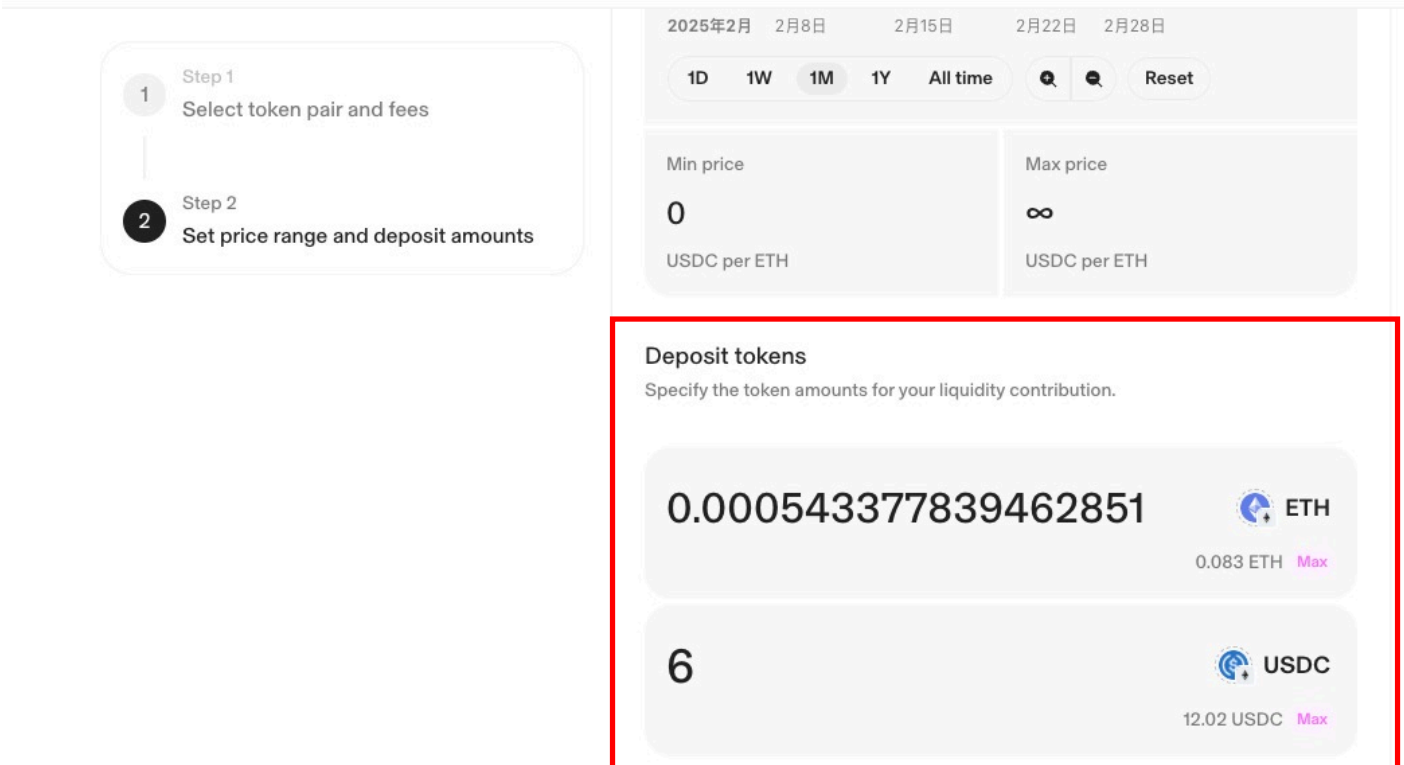
Interaction with Uniswap

- Set the price range to provide liquidity (full range or custom range can be selected).



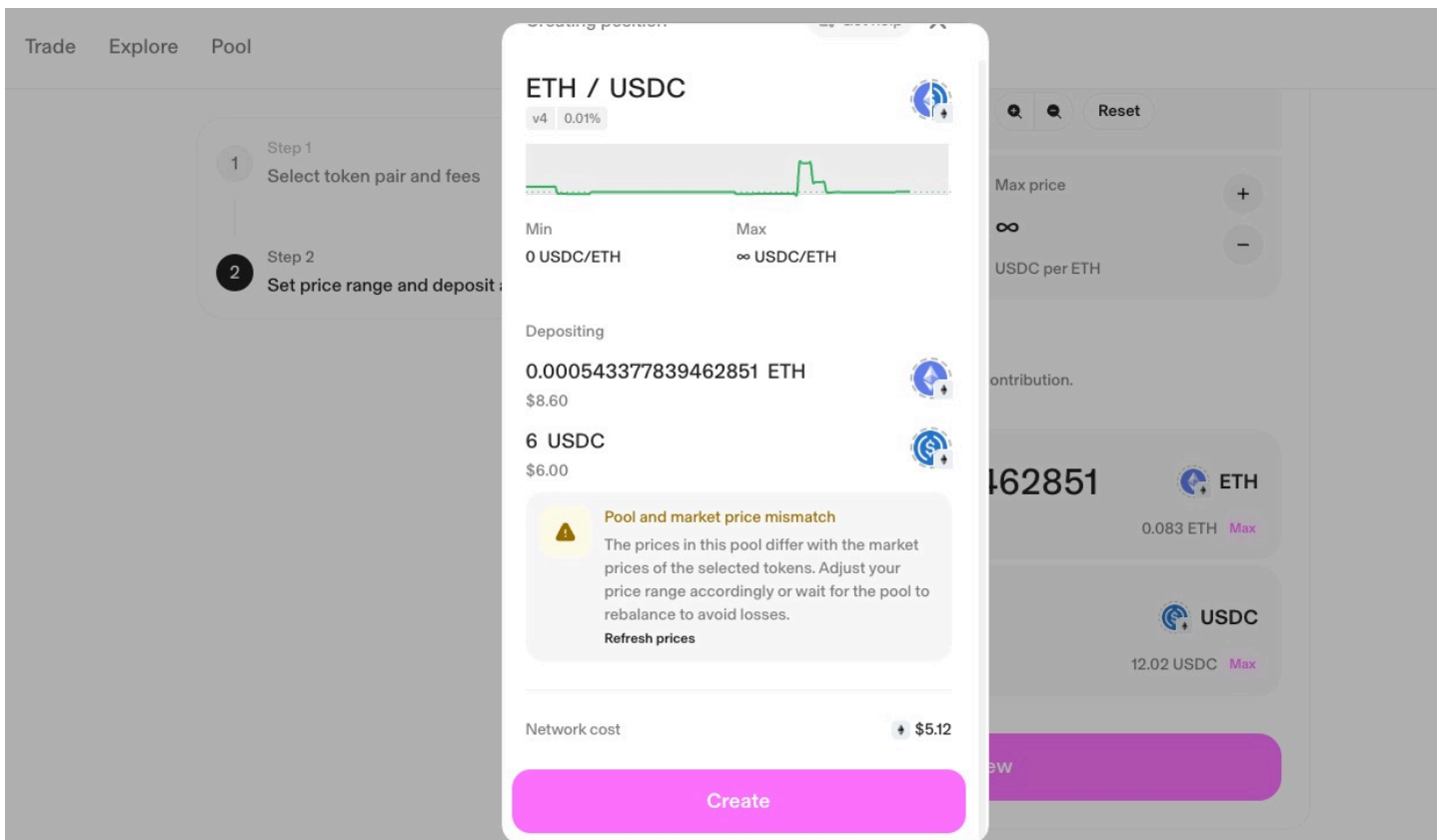
Interaction with Uniswap

- Enter the number of ETH and USDC you want to deposit.



Interaction with Uniswap

- Review the transaction details.





Interaction with Uniswap

- Go back to “Your positions” page to view your liquidity providing.

